LAB 8

TRANSFORMERS

This notebook trains a transformer encoder model on MagnaTagATune (MTAT) for multi-label music tagging. It is self-contained: installs dependencies, loads MTAT (local if available), trains, evaluates with ROC-AUC and PR-AUC, and reports compute/resource + emissions metrics.

URL for Google Colab: https://drive.google.com/file/d/17Gc6F2CCoc351pJ-oB6FAC-kLb6UY0UI/view?usp=sharing

URL for GitHub repo: https://github.com/manueleco/ml-for-sound-and-music/tree/main/lab08

URL for MTAT dataset in GoogleDrive: https://drive.google.com/drive/folders/1urEa6g919ewcRfHIU_RDX4B5yleuG67C?usp=sharing

SETUP

DEPENDENCIES

In [ ]:

```python
import sys, platform, os, time, math, random, shutil, subprocess

def in_colab():
    return "google.colab" in sys.modules

def pip_install(pkgs):
    subprocess.check_call([sys.executable, "-m", "pip", "install", "-q"] + pkgs)

# Install only on Colab
if in_colab():
    pip_install([
        "torch", "torchaudio",
        "tqdm", "scikit-learn", "matplotlib",
        "psutil", "codecarbon",
        "pandas", "numpy",
        "soundfile"
    ])
    # ffmpeg for mp3
    if shutil.which("ffmpeg") is None:
        subprocess.check_call(["apt-get", "update", "-y"])
        subprocess.check_call(["apt-get", "install", "-y", "ffmpeg"])
else:
    print("Local env -> skipping pip installs (use your venv).")

import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import torchaudio
import pandas as pd

from torch.utils.data import Dataset, DataLoader
from torchaudio.transforms import MelSpectrogram
from tqdm.auto import tqdm

from sklearn.metrics import roc_auc_score, average_precision_score
import psutil
from codecarbon import OfflineEmissionsTracker

class color:
   PURPLE = '\033[95m'
   CYAN = '\033[96m'
   DARKCYAN = '\033[36m'
   BLUE = '\033[94m'
   GREEN = '\033[92m'
   YELLOW = '\033[93m'
   RED = '\033[91m'
   BOLD = '\033[1m'
   UNDERLINE = '\033[4m'
   END = '\033[0m'

print("python:", sys.version)
print("platform:", platform.platform())
print("colab:", in_colab())
print("ffmpeg:", shutil.which("ffmpeg"))
print("torch:", torch.__version__)
print("torchaudio:", torchaudio.__version__)
```

```
Local env -> skipping pip installs (use your venv).
python: 3.11.14 (main, Oct  9 2025, 16:16:55) [Clang 17.0.0 (clang-1700.4.4.1)]
platform: macOS-26.1-arm64-arm-64bit
colab: False
ffmpeg: /opt/homebrew/bin/ffmpeg
torch: 2.9.1
torchaudio: 2.9.1
```

DEVICES (M1, CUDA OR CPU) & SCALABILITY

In [41]:

```python
SEED = 42
random.seed(SEED); np.random.seed(SEED); torch.manual_seed(SEED)

if torch.cuda.is_available():
    device = "cuda"
elif hasattr(torch.backends, "mps") and torch.backends.mps.is_available():
    # Mac M1
    device = "mps"
else:
    device = "cpu"
device
```

Out[41]:

```
'mps'
```

CONFIG

In [42]:

```python
CFG = {
    "mode": "tl",

    # audio
    "sample_rate": 16000,
    "clip_seconds": 2.0,
    "n_fft": 1024,
    "n_mels": 64,

    # training
    "batch_size": 32,
    "epochs": 10,
    "lr": 1e-3,

    # defaults (se sobreescriben abajo)
    "num_workers": 0,
    "pin_memory": False,

    # dataset
    "max_train_items": 8000,
    "max_val_items": 2000,
    "max_test_items": 2000,

    # baseline transformer
    "n_blocks": 2,
    "emb_dim": 64,
    "n_heads": 4,
    "stride": (4,4),
    "pos_enc_gain": 0.5,
    "cls_token_ratio": 0.25,

    # TL backbone
    "tl_backbone": "wav2vec2_base",
    "tl_freeze": True,
    "tl_pool": "mean",

    # early stop
    "early_stop_patience": 3
}

# smart dataloader settings (Colab + Drive vs Mac local)
if in_colab():
    # Drive + mp3 + workers suele colgarse entonces estable con 0
    CFG["num_workers"] = 0 if device != "cuda" else 2
else:
    # Local: MPS/CPU mejor 0, CUDA puede usar >0
    CFG["num_workers"] = 0 if device in ["mps", "cpu"] else 2

CFG["pin_memory"] = (device == "cuda")

CFG
```

Out[42]:

```
{'mode': 'tl',
 'sample_rate': 16000,
 'clip_seconds': 2.0,
 'n_fft': 1024,
 'n_mels': 64,
 'batch_size': 32,
 'epochs': 10,
 'lr': 0.001,
 'num_workers': 0,
 'pin_memory': False,
 'max_train_items': 8000,
 'max_val_items': 2000,
 'max_test_items': 2000,
 'n_blocks': 2,
 'emb_dim': 64,
 'n_heads': 4,
 'stride': (4, 4),
 'pos_enc_gain': 0.5,
 'cls_token_ratio': 0.25,
 'tl_backbone': 'wav2vec2_base',
 'tl_freeze': True,
 'tl_pool': 'mean',
 'early_stop_patience': 3}
```

In [43]:

```python
from pathlib import Path

def in_colab():
    return "google.colab" in sys.modules

def _has_mtat_layout(root: Path) -> bool:
    root = Path(root)
    return (
        (root / "annotations_final.csv").is_file()
        and (root / "clip_info_final.csv").is_file()
        and ((root / "audio").is_dir() or (root / "mp3").is_dir())
    )

def _mount_drive(mountpoint="/content/drive"):
    from google.colab import drive
    mp = Path(mountpoint)
    if mp.exists() and (mp / "MyDrive").exists():
        print("Drive already mounted:", mountpoint)
        return
    drive.mount(mountpoint)

def find_in_roots(roots, filename="annotations_final.csv"):
    for root in roots:
        root = Path(root)
        if not root.exists():
            continue
        for p in root.rglob(filename):
            cand = p.parent
            if _has_mtat_layout(cand):
                return cand
    return None

# Suggested default locations
if in_colab():
    _mount_drive("/content/drive")
    search_roots = [
        Path("/content/drive/MyDrive"),
        Path("/content/drive/MyDrive/data"),
        Path("/content"),
    ]
    MTAT_DATALABS_ROOT = find_in_roots(search_roots, "annotations_final.csv")
    if MTAT_DATALABS_ROOT is None:
        raise RuntimeError(
            "MTAT datalabs root not found in Google Drive.\n"
            "Expected a folder containing:\n"
            "  annotations_final.csv\n"
            "  clip_info_final.csv\n"
            "  audio/ (or mp3/)\n"
            "  please put your dataset under MyDrive/.../datalabs/"
        )
else:
    # local fallback
    MTAT_DATALABS_ROOT = None

print("MTAT_DATALABS_ROOT:", MTAT_DATALABS_ROOT)
```

```
MTAT_DATALABS_ROOT: None
```

LOAD MTAT

It'll take local route as prefered way.

Route resolver

In [44]:

```python
from pathlib import Path

# Optional local hint (not strictly needed because load_mtat() searches automatically but stil added)
LOCAL_CANDIDATES = [
    Path("data") / "mtat",
    Path("mtat"),
]

local_root = None
for p in LOCAL_CANDIDATES:
    if p.exists():
        local_root = p.resolve()
        break

print("Local candidate MTAT root (optional):", local_root)
```

```
Local candidate MTAT root (optional): None
```

MTAT Loader

In [ ]:

```python
import os, random
from pathlib import Path
import pandas as pd
import numpy as np

def read_csv_flexible(path):
    try:
        return pd.read_csv(path, sep="\t")
    except Exception:
        return pd.read_csv(path, sep=",")

def make_splits_from_ids(ids, split=(0.8, 0.1, 0.1), seed=SEED):
    ids = list(ids)
    rng = random.Random(seed)
    rng.shuffle(ids)
    n = len(ids)
    n_train = int(split[0] * n)
    n_val = int(split[1] * n)
    train_ids = ids[:n_train]
    val_ids = ids[n_train:n_train + n_val]
    test_ids = ids[n_train + n_val:]
    return train_ids, val_ids, test_ids

def find_datalabs_root():
    # If Colab and already found a valid root: use it
    if "google.colab" in sys.modules:
        if "MTAT_DATALABS_ROOT" in globals() and MTAT_DATALABS_ROOT is not None:
            return Path(MTAT_DATALABS_ROOT)

    anchors = [Path.cwd()]
    env_root = os.environ.get("MTAT_ROOT")
    if env_root:
        anchors.append(Path(env_root))

    for a in anchors:
        for p in [a] + list(a.parents):
            for cand in [
                p / "data" / "datalabs",
                p / "datalabs",
                p / "data" / "mtat",
                p / "mtat",
            ]:
                if (cand / "annotations_final.csv").exists() and (cand / "clip_info_final.csv").exists():
                    return cand
    raise FileNotFoundError("annotations_final.csv not found")

def resolve_audio_root(base):
    candidates = [
        base / "audio",
        base / "mp3",
        base / "audios",
        base.parent / "audio",
        base.parent / "mp3",
        base.parent / "audios",
    ]
    for c in candidates:
        if c.exists() and c.is_dir():
            return c
    raise FileNotFoundError("audio folder not found near datalabs/mtat root")

def normalize_mp3_rel(mp3_path):
    mp3_rel = str(mp3_path).strip()
    if mp3_rel.startswith("audio/"):
        mp3_rel = mp3_rel[len("audio/"):]
    if mp3_rel.startswith("./"):
        mp3_rel = mp3_rel[2:]
    return mp3_rel

def load_mtat():
    base = find_datalabs_root()
    ann_path = base / "annotations_final.csv"
    info_path = base / "clip_info_final.csv"
    audio_root = resolve_audio_root(base)

    df_ann = read_csv_flexible(ann_path)
    df_info = read_csv_flexible(info_path)

    df_ann.columns = [c.strip() for c in df_ann.columns]
    df_info.columns = [c.strip() for c in df_info.columns]

    tag_cols = [c for c in df_ann.columns if c != "clip_id" and df_ann[c].dtype != object]
    ann_idx = df_ann.set_index("clip_id")

    info_idx = df_info.set_index("clip_id")

    all_ids = df_ann["clip_id"].values
    train_ids, val_ids, test_ids = make_splits_from_ids(all_ids)

    def build_df(id_list):
        rows = []
        for cid in id_list:
            if cid in info_idx.index:
                mp3_rel = normalize_mp3_rel(info_idx.loc[cid, "mp3_path"])
                rows.append({"clip_id": cid, "mp3_path": mp3_rel})
        return pd.DataFrame(rows)

    train_df = build_df(train_ids).reset_index(drop=True)
    val_df = build_df(val_ids).reset_index(drop=True)
    test_df = build_df(test_ids).reset_index(drop=True)

    def filter_existing(df):
        paths = (df["mp3_path"].astype(str)).tolist()
        exists = [(audio_root / p).is_file() for p in paths]
        return df.loc[exists].reset_index(drop=True)

    train_df = filter_existing(train_df)
    val_df = filter_existing(val_df)
    test_df = filter_existing(test_df)

    return ann_idx, tag_cols, audio_root, train_df, val_df, test_df

ann_idx, TAG_COLS, AUDIO_ROOT, train_df, val_df, test_df = load_mtat()
```

SUBSAMPLE

In [46]:

```python
def subsample_df(df, max_items):
    if (max_items is None) or (len(df) <= max_items):
        return df
    idx = np.random.RandomState(SEED).choice(len(df), size=max_items, replace=False)
    return df.iloc[sorted(idx)].reset_index(drop=True)

train_df = subsample_df(train_df, CFG["max_train_items"])
val_df = subsample_df(val_df, CFG["max_val_items"])
test_df = subsample_df(test_df, CFG["max_test_items"])

len(train_df), len(val_df), len(test_df)
```

Out[46]:

```
(8000, 2000, 2000)
```

LABELING

In [47]:

```python
n_classes = len(TAG_COLS)
n_classes, TAG_COLS[:10]
```

Out[47]:

```
(188,
 ['no voice',
  'singer',
  'duet',
  'plucking',
  'hard rock',
  'world',
  'bongos',
  'harpsichord',
  'female singing',
  'clasical'])
```

DATASET AND FEATURE EXTRACTOR

Dataset

In [48]:

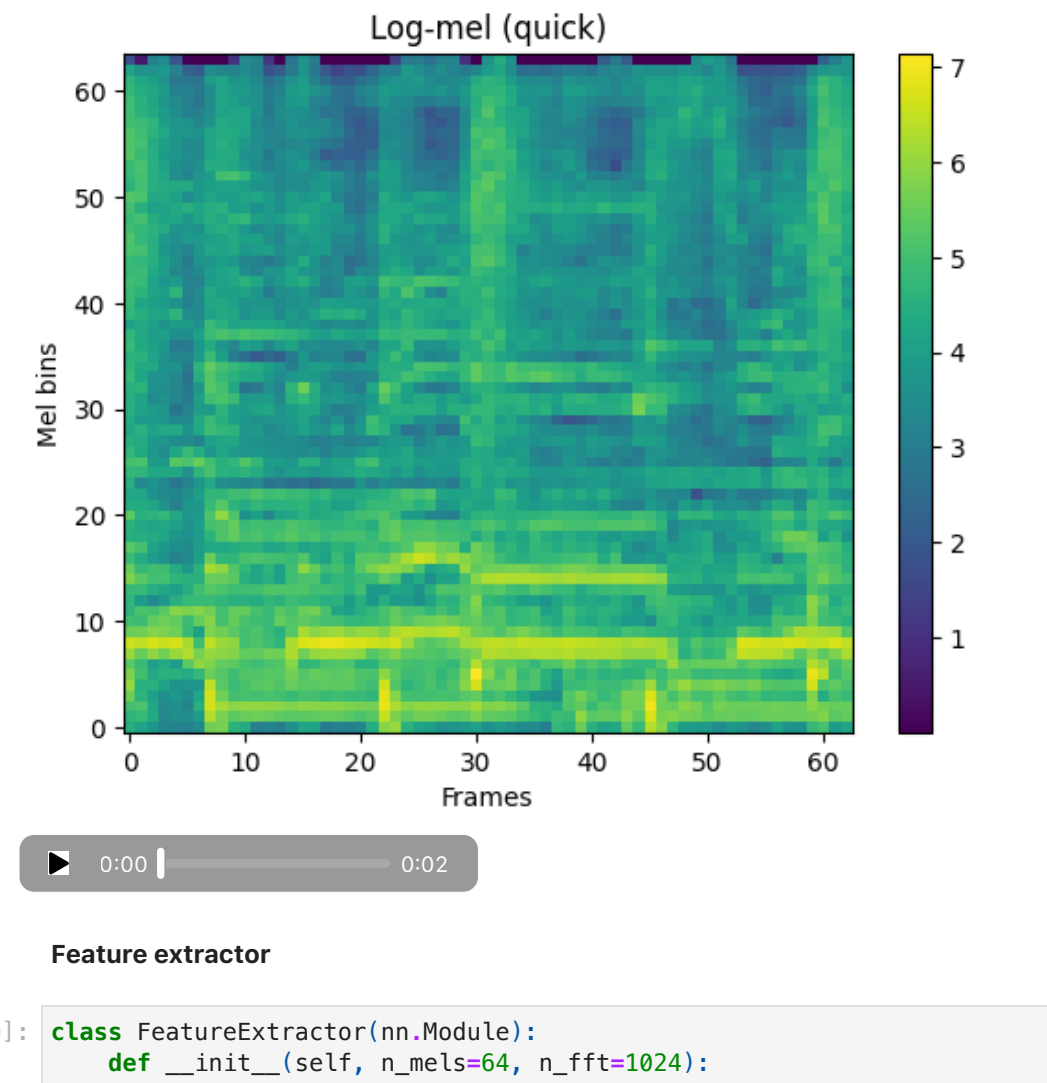
```python
import numpy as np, subprocess, shutil

def _ffmpeg_decode_to_float32_mono(path, target_sr=16000):
    ff = shutil.which("ffmpeg")
    if ff is None:
        raise RuntimeError("ffmpeg not found in PATH")
    cmd = [
        ff, "-nostdin", "-v", "error",
        "-i", str(path),
        "-ac", "1",
        "-ar", str(int(target_sr)),
        "-f", "f32le",
        "pipe:1",
    ]
    raw = subprocess.check_output(cmd, stderr=subprocess.STDOUT)
    y = np.frombuffer(raw, dtype=np.float32)
    if y.size == 0:
        raise RuntimeError(f"ffmpeg decoded empty audio: {path}")
    return torch.from_numpy(y)

class MTATDataset(Dataset):
    """
    Robust MTAT dataset:
    + Filters out missing/bad audio paths at init time.
    + Loads with torchaudio when possible.
    + Falls back to ffmpeg when torchaudio fails on some MP3s.
    + Retries on occasional bad files to avoid crashing training.
    """
    def __init__(
        self, df, ann_idx, tag_cols, audio_root,
        sample_rate=16000, clip_seconds=2.0,
        device="cpu", split="train",
        max_retries=8
    ):
        self.audio_root = Path(audio_root)
        self.ann_idx = ann_idx
        self.tag_cols = list(tag_cols)
        self.sample_rate = int(sample_rate)
        self.n_samples = int(sample_rate * clip_seconds)
        self.device = device
        self.split = split
        self.max_retries = int(max_retries)

        # Make a clean copy
        df = df.reset_index(drop=True).copy()
        df["mp3_path"] = df["mp3_path"].astype(str).str.strip()

        # Filter rows whose audio file exists & is a regular file
        ok = []
        for p in df["mp3_path"].tolist():
            fp = self.audio_root / p
            ok.append(fp.is_file())
        self.df = df.loc[ok].reset_index(drop=True)

        if len(self.df) == 0:
            raise RuntimeError(
                f"MTATDataset has 0 valid audio files. "
                f"audio_root={self.audio_root} (check folder + mp3_path format)"
            )

    def __len__(self):
        return len(self.df)

    def _pad_or_crop(self, x):
        if x.numel() < self.n_samples:
            x = F.pad(x, (0, self.n_samples - x.numel()))
        elif x.numel() > self.n_samples:
            if self.split == "train":
                start = random.randint(0, x.numel() - self.n_samples)
            else:
                start = (x.numel() - self.n_samples) // 2
            x = x[start:start + self.n_samples]
        return x

    def _load_audio(self, path: Path):
        # try torchaudio (fast when it works)
        try:
            wav, sr = torchaudio.load(str(path))
            if wav.dim() == 2:
                # mono
                wav = wav.mean(dim=0)
            if sr != self.sample_rate:
                wav = torchaudio.functional.resample(wav, sr, self.sample_rate)
            return wav.contiguous()
        except Exception:
            # fallback: ffmpeg (handles nasty MP3s)
            y = _ffmpeg_decode_to_float32_mono(path, target_sr=self.sample_rate)
            return y.contiguous()

    def __getitem__(self, idx):
        for _ in range(self.max_retries):
            row = self.df.iloc[int(idx)]
            clip_id = row["clip_id"]
            mp3_rel = str(row["mp3_path"]).strip()
            path = self.audio_root / mp3_rel

            try:
                x = self._load_audio(path)
                x = self._pad_or_crop(x)

                y = self.ann_idx.loc[clip_id, self.tag_cols].values.astype(np.float32)
                y = torch.tensor(y, dtype=torch.float32)

                return {"audio": x.to(self.device), "labels": y.to(self.device)}

            except Exception:
                idx = random.randrange(len(self.df))

        # hard fallback (never crash training)
        row = self.df.iloc[0]
        clip_id = row["clip_id"]
        y = self.ann_idx.loc[clip_id, self.tag_cols].values.astype(np.float32)
        y = torch.tensor(y, dtype=torch.float32)
        x = torch.zeros(self.n_samples, dtype=torch.float32)
        return {"audio": x.to(self.device), "labels": y.to(self.device)}

train_dataset = MTATDataset(train_df, ann_idx, TAG_COLS, AUDIO_ROOT, CFG["sample_rate"], CFG["clip_seconds"], device=device, split="train")
val_dataset   = MTATDataset(val_df,   ann_idx, TAG_COLS, AUDIO_ROOT, CFG["sample_rate"], CFG["clip_seconds"], device=device, split="val")
test_dataset  = MTATDataset(test_df,  ann_idx, TAG_COLS, AUDIO_ROOT, CFG["sample_rate"], CFG["clip_seconds"], device=device, split="test")

train_loader = DataLoader(
    train_dataset,
    batch_size=CFG["batch_size"],
    shuffle=True,
    num_workers=CFG["num_workers"],
    pin_memory=CFG["pin_memory"],
    drop_last=True
)
val_loader   = DataLoader(
    val_dataset,
    batch_size=CFG["batch_size"],
    shuffle=False,
    num_workers=CFG["num_workers"],
    pin_memory=CFG["pin_memory"]
)
test_loader  = DataLoader(
    test_dataset,
    batch_size=CFG["batch_size"],
    shuffle=False,
    num_workers=CFG["num_workers"],
    pin_memory=CFG["pin_memory"]
)

print("Valid items:", len(train_dataset), len(val_dataset), len(test_dataset))
```

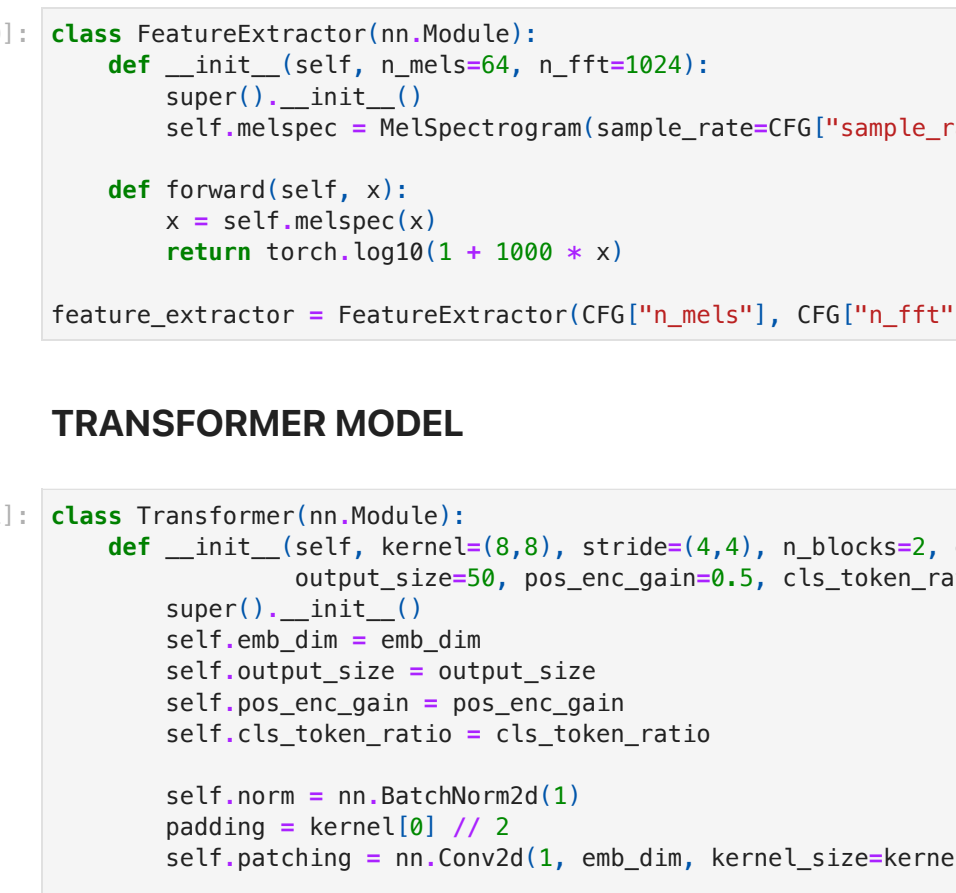
```
Valid items: 8000 2000 2000
```

Audio visualization waveform mel and playback

In [49]:

```python
import matplotlib.pyplot as plt

try:
    from IPython.display import Audio, display
except Exception:
    Audio = None
    display = None

sample = train_dataset[0]
x = sample["audio"].detach().cpu()

plt.figure()
plt.plot(x.numpy())
plt.title("Waveform (2s clip)")
plt.xlabel("Samples")
plt.ylabel("Amplitude")
plt.show()

melspec_tmp = MelSpectrogram(sample_rate=CFG["sample_rate"], n_mels=CFG["n_mels"], n_fft=CFG["n_fft"])
mel = torch.log10(1 + 1000 * melspec_tmp(x.unsqueeze(0))).squeeze(0).numpy()

plt.figure()
plt.imshow(mel, aspect="auto", origin="lower")
plt.title("Log-mel (quick)")
plt.xlabel("Frames")
plt.ylabel("Mel bins")
plt.colorbar()
plt.show()

if Audio is not None and display is not None:
    display(Audio(x.numpy(), rate=CFG["sample_rate"]))
```





0:00 0:02

Feature extractor

In [50]:

```python
class FeatureExtractor(nn.Module):
    def __init__(self, n_mels=64, n_fft=1024):
        super().__init__()
        self.melspec = MelSpectrogram(sample_rate=CFG["sample_rate"], n_mels=n_mels, n_fft=n_fft)

    def forward(self, x):
        x = self.melspec(x)
        return torch.log10(1 + 1000 * x)

feature_extractor = FeatureExtractor(CFG["n_mels"], CFG["n_fft"]).to(device)
```

TRANSFORMER MODEL

In [51]:

```python
class Transformer(nn.Module):
    def __init__(self, kernel=(8,8), stride=(4,4), n_blocks=2, emb_dim=64, n_heads=4, mlp_ratio=2,
                 output_size=50, pos_enc_gain=0.5, cls_token_ratio=0.25):
        super().__init__()
        self.emb_dim = emb_dim
        self.output_size = output_size
        self.pos_enc_gain = pos_enc_gain
        self.cls_token_ratio = cls_token_ratio

        self.norm = nn.BatchNorm2d(1)
        padding = kernel[0] // 2
        self.patching = nn.Conv2d(1, emb_dim, kernel_size=kernel, stride=stride, padding=padding)

        self.blocks = nn.ModuleList([
            nn.TransformerEncoderLayer(
                emb_dim, n_heads,
                dim_feedforward=emb_dim * mlp_ratio,
                dropout=0.1,
                norm_first=True
            ) for _ in range(n_blocks)
        ])

        self.cls = nn.Parameter(torch.randn(1,1,emb_dim))
        nn.init.normal_(self.cls, std=0.02)

        self.fc1 = nn.Linear(emb_dim, emb_dim)
        self.fc2 = nn.Linear(emb_dim, output_size)

    def positional_encoding(self, seq_len):
        position = torch.arange(seq_len, dtype=torch.float32).unsqueeze(1)
        div_term = torch.exp(torch.arange(0, self.emb_dim, 2, dtype=torch.float32) * -(math.log(10000.0) / self.emb_dim))
        pe = torch.zeros(seq_len, self.emb_dim, device=self.cls.device)
        pe[:, 0::2] = torch.sin(position * div_term)
        pe[:, 1::2] = torch.cos(position * div_term)
        return pe

    def forward(self, x):
        x = x.unsqueeze(1)
        x = self.norm(x)
        x = self.patching(x)
        B, E, M, T = x.shape

        pe_m = self.positional_encoding(M)
        pe_t = self.positional_encoding(T)
        pe_m = pe_m.transpose(0,1).unsqueeze(0).unsqueeze(3)
        pe_t = pe_t.transpose(0,1).unsqueeze(0).unsqueeze(2)
        x = x + self.pos_enc_gain * (pe_m + pe_t)

        x = x.flatten(start_dim=2).swapaxes(1,2)
        cls = self.cls.repeat(B,1,1)
        x = torch.cat([cls, x], dim=1)

        for blk in self.blocks:
            x = blk(x)

        x_cls = x[:,0,:]
        x_mean = x[:,1:,:].mean(dim=1)
        x = self.cls_token_ratio * x_cls + (1 - self.cls_token_ratio) * x_mean

        x = F.layer_norm(x, (self.emb_dim,))
        x = F.relu(self.fc1(x))
        logits = self.fc2(x)
        return logits


class TLWav2Vec2Tagger(nn.Module):
    def __init__(self, n_classes, sample_rate=16000, backbone_name="wav2vec2_base", freeze=True, pool="mean"):
        super().__init__()
        if backbone_name == "wav2vec2_base":
            bundle = torchaudio.pipelines.WAV2VEC2_BASE
        elif backbone_name == "wav2vec2_base_960h":
            bundle = torchaudio.pipelines.WAV2VEC2_BASE_960H
        else:
            raise ValueError(f"Unknown tl_backbone: {backbone_name}")

        self.sample_rate = int(sample_rate)
        self.pool = str(pool)
        self.backbone = bundle.get_model()

        if freeze:
            for p in self.backbone.parameters():
                p.requires_grad = False

        self.emb_dim = int(bundle._params["encoder_embed_dim"])
        self.head = nn.Linear(self.emb_dim, int(n_classes))

    def forward(self, x):
        if x.dim() == 1:
            x = x.unsqueeze(0)
        feats, _ = self.backbone.extract_features(x)
        z = feats[-1]
        if self.pool == "mean":
            z = z.mean(dim=1)
        elif self.pool == "max":
            z = z.amax(dim=1)
        else:
            raise ValueError(f"Unknown tl_pool: {self.pool}")
        logits = self.head(z)
        return logits



# Models are buit later (TL + baseline). Basically for a fair comparison.
# This cell only defines the model classes.
# Commented section because redundant and cause a lot of workload in the CPU and it's time consuming.
'''
if CFG["mode"] == "baseline":
    model = Transformer(
        n_blocks=CFG["n_blocks"],
        emb_dim=CFG["emb_dim"],
        n_heads=CFG["n_heads"],
        stride=CFG["stride"],
        pos_enc_gain=CFG["pos_enc_gain"],
        cls_token_ratio=CFG["cls_token_ratio"],
        output_size=n_classes
    ).to(device)
else:
    model = TLWav2Vec2Tagger(
        n_classes=n_classes,
        sample_rate=CFG["sample_rate"],
        backbone_name=CFG["tl_backbone"],
        freeze=CFG["tl_freeze"],
        pool=CFG["tl_pool"],
    ).to(device)

sum(p.numel() for p in model.parameters() if p.requires_grad)
'''
```

Out[51]:

```
'\nif CFG["mode"] == "baseline":\n    model = Transformer(\n        n_blocks=CFG["n_blocks"],\n        emb_dim=CFG["emb_dim"],\n        n_heads=CFG["n_heads"],\n        stride=CFG["stride"],\
n        pos_enc_gain=CFG["pos_enc_gain"],\n        cls_token_ratio=CFG["cls_token_ratio"],\n        output_size=n_classes\n    ).to(device)\nelse:\n    model = TLWav2Vec2Tagger(\n        n_c
lasses=n_classes,\n        sample_rate=CFG["sample_rate"],\n        backbone_name=CFG["tl_backbone"],\n        freeze=CFG["tl_freeze"],\n        pool=CFG["tl_pool"],\n    ).to(device)\n\nsum(
p.numel() for p in model.parameters() if p.requires_grad)\n'
```

RESOURCE MONITOR AND ENVIRONMENTAL IMPACT

Resource monitor (CPU, RAM and time)

In [ ]:

```python
class ResourceMonitor:

    def __init__(self, interval_s=1.0):
        self.interval_s = interval_s
        self.proc = psutil.Process(os.getpid())
        self.running = False
        self.samples = []
        self._t0 = None
        self._thread = None

    def _get_accel_mem_mb(self):
        # CUDA
        if torch.cuda.is_available():
```

```
        if torch.cuda.is_available():
            try:
                return float(torch.cuda.memory_allocated() / (1024**2))
            except Exception:
                return None
        # MPS (Apple Silicon or M1 technically)
        if hasattr(torch, "mps") and torch.backends.mps.is_available():
            try:
                if hasattr(torch.mps, "current_allocated_memory"):
                    return float(torch.mps.current_allocated_memory() / (1024**2))
            except Exception:
                return None
        return None

    def start(self):
        import threading
        self.running = True
        self._t0 = time.time()

        def loop():
            while self.running:
                t = time.time() - self._t0
                cpu = psutil.cpu_percent(interval=None)
                ram_mb = self.proc.memory_info().rss / (1024**2)
                accel_mb = self._get_accel_mem_mb()
                self.samples.append((t, cpu, ram_mb, accel_mb))
                time.sleep(self.interval_s)

        self._thread = threading.Thread(target=loop, daemon=True)
        self._thread.start()

    def stop(self):
        self.running = False
        time.sleep(self.interval_s)

    def summary(self):
        if not self.samples:
            return {}
        arr = np.array([[s[0], s[1], s[2], (s[3] if s[3] is not None else np.nan)] for s in self.samples], dtype=float)
        out = {
            "wall_time_s": float(arr[-1, 0]),
            "cpu_mean_pct": float(np.nanmean(arr[:, 1])),
            "cpu_max_pct": float(np.nanmax(arr[:, 1])),
            "ram_mean_mb": float(np.nanmean(arr[:, 2])),
            "ram_max_mb": float(np.nanmax(arr[:, 2])),
            "n_samples": int(len(self.samples)),
        }
        if not np.all(np.isnan(arr[:, 3])):
            out.update({
                "accel_mem_mean_mb": float(np.nanmean(arr[:, 3])),
                "accel_mem_max_mb": float(np.nanmax(arr[:, 3])),
            })
        return out
```

ROC AUC AND PR AUC METRICS

ROC-AUD and PR-AUC metrics plus fast training with early stop.

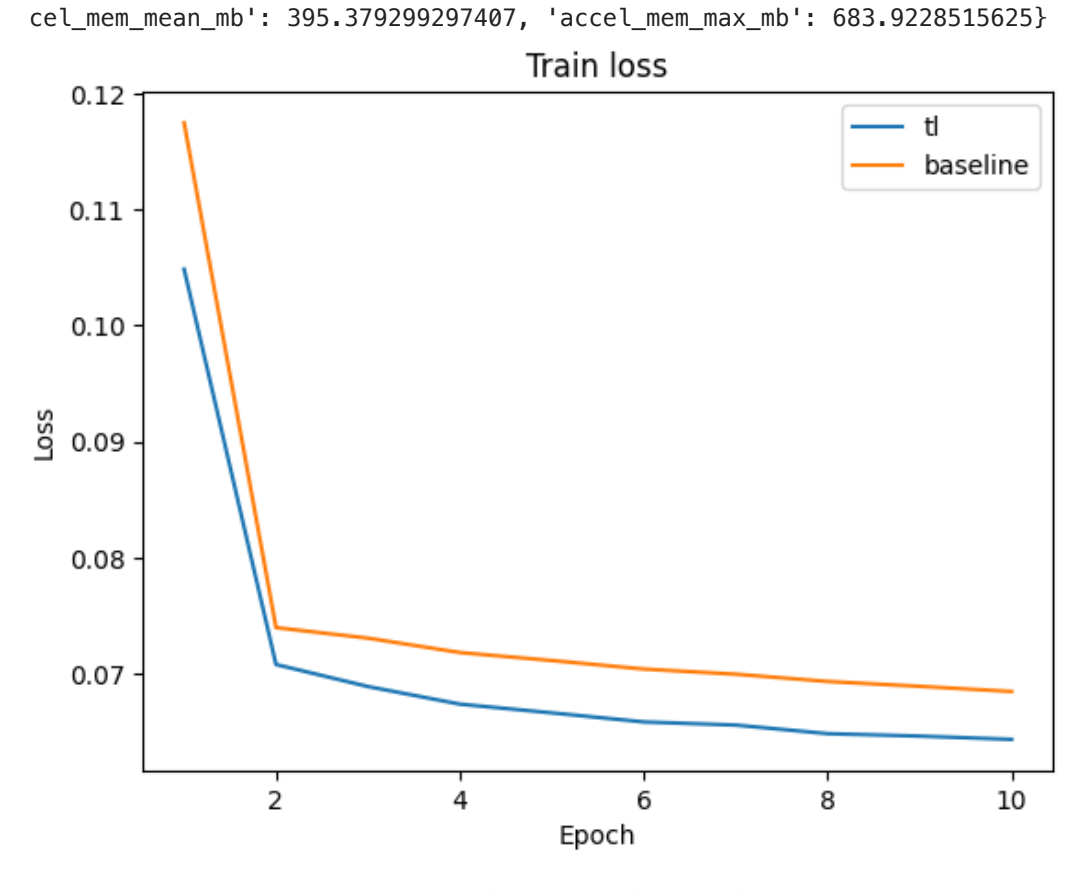
In [53]:

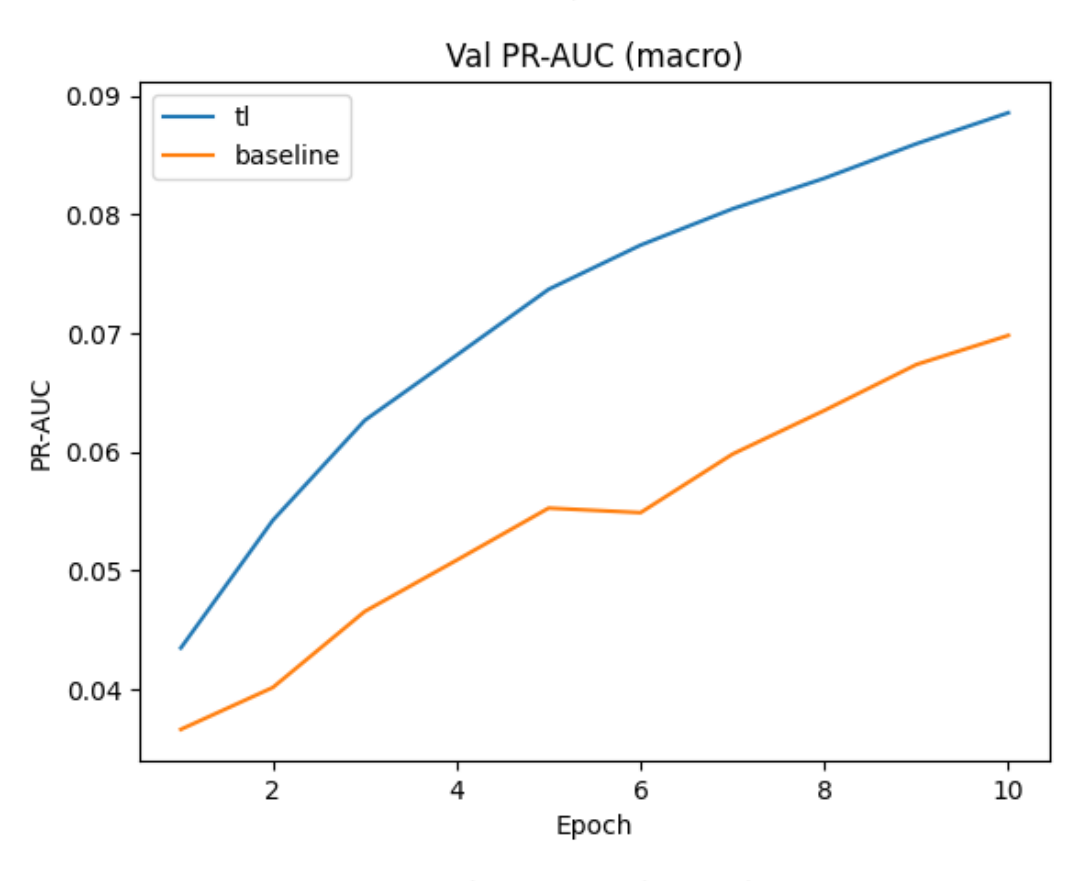
```
def forward_batch(model, batch, mode):
    x = batch["audio"].to(device)
    y = batch["labels"].to(device)

    if mode == "baseline":
        x_in = feature_extractor(x)
        logits = model(x_in)
    else:
        logits = model(x)

    probs = torch.sigmoid(logits)
    return logits, probs, y

def eval_auc(model, loader, mode):
    model.eval()
    ys, ps = [], []
    with torch.no_grad():
        for batch in loader:
            _, probs, y = forward_batch(model, batch, mode)
            ys.append(y.detach().cpu().numpy())
            ps.append(probs.detach().cpu().numpy())

    y_true = np.concatenate(ys, axis=0)
    y_score = np.concatenate(ps, axis=0)

    pos = y_true.sum(axis=0)
    n = y_true.shape[0]
    valid = (pos > 0) & (pos < n)

    n_valid = int(valid.sum())
    n_total = int(y_true.shape[1])

    if n_valid == 0:
        return float("nan"), float("nan"), {"valid_classes": 0, "total_classes": n_total, "dropped_classes": n_total}

    y_true_v = y_true[:, valid]
    y_score_v = y_score[:, valid]

    roc = roc_auc_score(y_true_v, y_score_v, average="macro")
    pr  = average_precision_score(y_true_v, y_score_v, average="macro")

    diag = {
        "valid_classes": n_valid,
        "total_classes": n_total,
        "dropped_classes": n_total - n_valid
    }
    return float(roc), float(pr), diag

def train_loop(model, mode, run_name="run"):
    os.makedirs("emissions", exist_ok=True)
    tracker = OfflineEmissionsTracker(
        country_iso_code="ESP",
        output_dir="emissions",
        output_file=f"codecarbon_emissions_{run_name}.csv",
        log_level="error"
    )
    monitor = ResourceMonitor(interval_s=1.0)

    optimizer = torch.optim.Adam(model.parameters(), lr=CFG["lr"])
    loss_fn = nn.BCEWithLogitsLoss()

    best_pr = -1.0
    bad_epochs = 0
    history = []
    best_state = None

    tracker.start()
    monitor.start()

    try:
        for epoch in range(CFG["epochs"]):
            t0 = time.time()
            model.train()
            total_loss = 0.0

            pbar = tqdm(train_loader, desc=f"{run_name} | Epoch {epoch+1}/{CFG['epochs']}", leave=True)
            for step, batch in enumerate(pbar, start=1):
                logits, _, y = forward_batch(model, batch, mode)
                loss = loss_fn(logits, y)

                optimizer.zero_grad()
                loss.backward()
                optimizer.step()

                total_loss += loss.item()
                pbar.set_postfix(loss=f"{loss.item():.4f}", avg=f"{(total_loss/step):.4f}")

            train_loss = total_loss / max(1, len(train_loader))
            val_roc, val_pr, val_diag = eval_auc(model, val_loader, mode)
            dt = time.time() - t0

            history.append({
                "epoch": epoch+1,
                "train_loss": float(train_loss),
                "val_roc": float(val_roc),
                "val_pr": float(val_pr),
                "epoch_time_s": float(dt),
                "val_valid_tags": int(val_diag["valid_classes"]),
                "val_total_tags": int(val_diag["total_classes"]),
                "val_dropped_tags": int(val_diag["dropped_classes"]),
            })

            if val_pr > best_pr + 1e-4:
                best_pr = float(val_pr)
                bad_epochs = 0
                best_state = {k: v.detach().cpu().clone() for k, v in model.state_dict().items()}
            else:
                bad_epochs += 1
                if bad_epochs >= CFG["early_stop_patience"]:
                    print(f"{run_name} early stop at epoch {epoch+1} (best val PR-AUC={best_pr:.4f})")
                    break

            print(
                f"{run_name} | epoch {epoch+1:02d} | time={dt:.1f}s | train_loss={train_loss:.4f} | "
                f"val_ROC-AUC={val_roc:.4f} | val_PR-AUC={val_pr:.4f} | "
                f"valid_tags={val_diag['valid_classes']}/{val_diag['total_classes']}"
            )

        if best_state is not None:
            model.load_state_dict(best_state)

    finally:
        monitor.stop()
        emissions_kg = tracker.stop()
        resources = monitor.summary()

    test_roc, test_pr, test_diag = eval_auc(model, test_loader, mode)

    out = {
        "history": history,
        "test_roc": float(test_roc),
        "test_pr": float(test_pr),
        "test_diag": test_diag,
        "emissions_kg": emissions_kg,
        "resources": resources,
    }
    return out
```

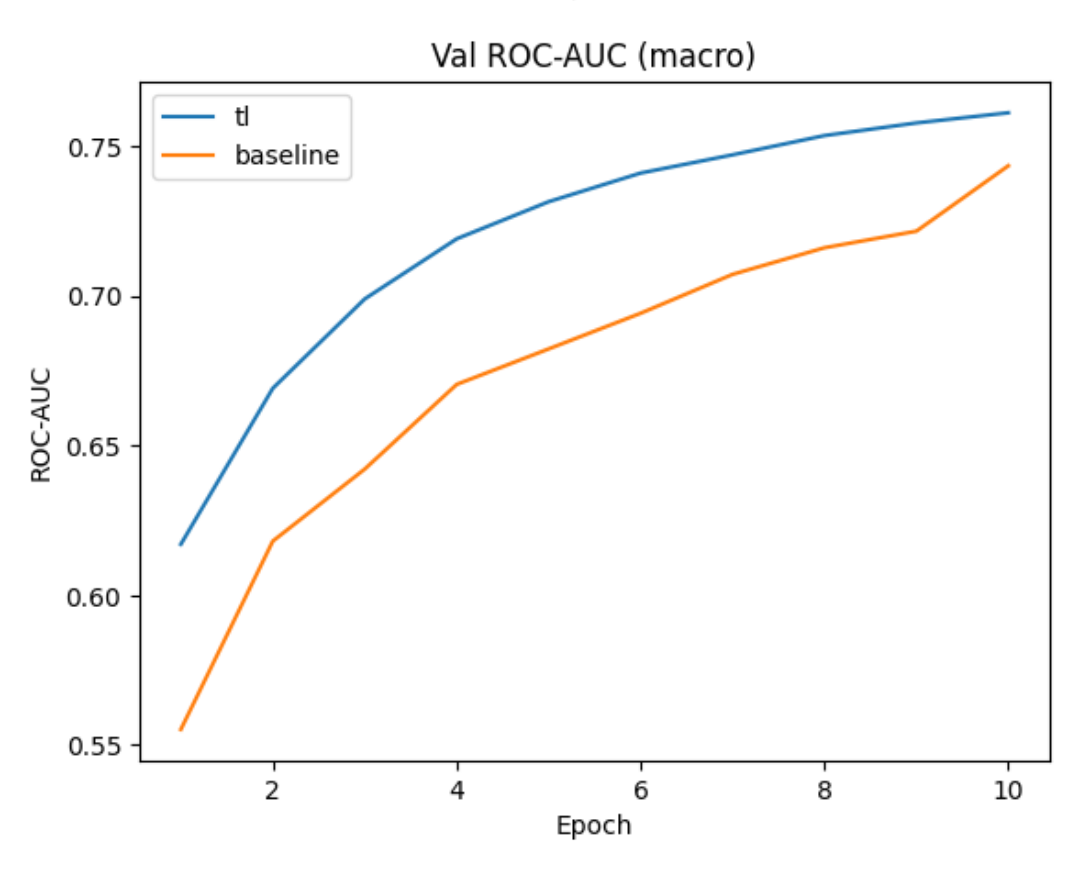
In [54]:

```
def build_model(mode):
    if mode == "baseline":
        return Transformer(
            n_blocks=CFG["n_blocks"],
            emb_dim=CFG["emb_dim"],
            n_heads=CFG["n_heads"],
            stride=CFG["stride"],
            pos_enc_gain=CFG["pos_enc_gain"],
            cls_token_ratio=CFG["cls_token_ratio"],
            output_size=n_classes
        ).to(device)
    else:
        return TLWav2Vec2Tagger(
            n_classes=n_classes,
            sample_rate=CFG["sample_rate"],
            backbone_name=CFG["tl_backbone"],
            freeze=CFG["tl_freeze"],
            pool=CFG["tl_pool"],
        ).to(device)

results = {}
trained_models = {}  # additional: keep trained models for final sanity checks..

for mode in ["tl", "baseline"]:
    m = build_model(mode)
    run_name = mode

    res = train_loop(m, mode=mode, run_name=run_name)

    # addi: store the trained model (already loaded with best_state inside train_loop)
    m.eval()
    trained_models[mode] = m

    trainable = sum(p.numel() for p in m.parameters() if p.requires_grad)
    total = sum(p.numel() for p in m.parameters())
    print(color.BOLD + f"\n{mode.upper()} params" + color.END)
    print(f"Trainable params: {trainable:,} / Total params: {total:,}")
    results[mode] = res

    print(color.BOLD + f"\n{mode.upper()} SUMMARY" + color.END)
    print(f"TEST ROC-AUC (macro): {res['test_roc']:.4f}")
    print(f"TEST PR-AUC  (macro): {res['test_pr']:.4f}")
    print(f"TEST valid_tags: {res['test_diag']['valid_classes']}/{res['test_diag']['total_classes']} (dropped={res['test_diag']['dropped_classes']})")
    print("Emissions (kg CO2eq):", res["emissions_kg"])
    print("Resources summary:", res["resources"])

import pandas as pd
import matplotlib.pyplot as plt

df_tl = pd.DataFrame(results["tl"]["history"])
df_bl = pd.DataFrame(results["baseline"]["history"])

plt.figure()
plt.plot(df_tl["epoch"], df_tl["train_loss"], label="tl")
plt.plot(df_bl["epoch"], df_bl["train_loss"], label="baseline")
plt.title("Train loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.show()

plt.figure()
plt.plot(df_tl["epoch"], df_tl["val_pr"], label="tl")
plt.plot(df_bl["epoch"], df_bl["val_pr"], label="baseline")
plt.title("Val PR-AUC (macro)")
plt.xlabel("Epoch")
plt.ylabel("PR-AUC")
plt.legend()
plt.show()

plt.figure()
plt.plot(df_tl["epoch"], df_tl["val_roc"], label="tl")
plt.plot(df_bl["epoch"], df_bl["val_roc"], label="baseline")
plt.title("Val ROC-AUC (macro)")
plt.xlabel("Epoch")
plt.ylabel("ROC-AUC")
plt.legend()
plt.show()
```

```
tl | Epoch 1/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 01 | time=238.9s | train_loss=0.1049 | val_ROC-AUC=0.6170 | val_PR-AUC=0.0435 | valid_tags=186/188
tl | Epoch 2/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 02 | time=244.0s | train_loss=0.0708 | val_ROC-AUC=0.6691 | val_PR-AUC=0.0542 | valid_tags=186/188
tl | Epoch 3/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 03 | time=246.1s | train_loss=0.0689 | val_ROC-AUC=0.6989 | val_PR-AUC=0.0626 | valid_tags=186/188
tl | Epoch 4/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 04 | time=247.0s | train_loss=0.0674 | val_ROC-AUC=0.7189 | val_PR-AUC=0.0681 | valid_tags=186/188
tl | Epoch 5/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 05 | time=245.7s | train_loss=0.0667 | val_ROC-AUC=0.7313 | val_PR-AUC=0.0737 | valid_tags=186/188
tl | Epoch 6/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 06 | time=242.0s | train_loss=0.0659 | val_ROC-AUC=0.7409 | val_PR-AUC=0.0774 | valid_tags=186/188
tl | Epoch 7/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 07 | time=243.2s | train_loss=0.0656 | val_ROC-AUC=0.7470 | val_PR-AUC=0.0805 | valid_tags=186/188
tl | Epoch 8/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 08 | time=244.3s | train_loss=0.0649 | val_ROC-AUC=0.7534 | val_PR-AUC=0.0830 | valid_tags=186/188
tl | Epoch 9/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 09 | time=239.0s | train_loss=0.0647 | val_ROC-AUC=0.7576 | val_PR-AUC=0.0859 | valid_tags=186/188
tl | Epoch 10/10:   0%|          | 0/250 [00:00<?, ?it/s]
tl | epoch 10 | time=237.9s | train_loss=0.0644 | val_ROC-AUC=0.7610 | val_PR-AUC=0.0886 | valid_tags=186/188

TL params
Trainable params: 144,572 / Total params: 94,515,516

TL SUMMARY
TEST ROC-AUC (macro): 0.7621
TEST PR-AUC  (macro): 0.0868
TEST valid_tags: 188/188 (dropped=0)
Emissions (kg CO2eq): 0.001293008618316594
Resources summary: {'wall_time_s': 2430.143227815628, 'cpu_mean_pct': 23.911890999174236, 'cpu_max_pct': 90.1, 'ram_mean_mb': 815.8569170623451, 'ram_max_mb': 1159.28125, 'n_samples': 2422, 'ac
cel_mem_mean_mb': 406.30812724181976, 'accel_mem_max_mb': 785.61328125}
baseline | Epoch 1/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 01 | time=105.3s | train_loss=0.1175 | val_ROC-AUC=0.5552 | val_PR-AUC=0.0366 | valid_tags=186/188
baseline | Epoch 2/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 02 | time=104.4s | train_loss=0.0740 | val_ROC-AUC=0.6180 | val_PR-AUC=0.0401 | valid_tags=186/188
baseline | Epoch 3/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 03 | time=107.8s | train_loss=0.0731 | val_ROC-AUC=0.6421 | val_PR-AUC=0.0465 | valid_tags=186/188
baseline | Epoch 4/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 04 | time=104.5s | train_loss=0.0718 | val_ROC-AUC=0.6703 | val_PR-AUC=0.0509 | valid_tags=186/188
baseline | Epoch 5/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 05 | time=103.7s | train_loss=0.0712 | val_ROC-AUC=0.6822 | val_PR-AUC=0.0552 | valid_tags=186/188
baseline | Epoch 6/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 06 | time=105.0s | train_loss=0.0704 | val_ROC-AUC=0.6941 | val_PR-AUC=0.0549 | valid_tags=186/188
baseline | Epoch 7/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 07 | time=100.1s | train_loss=0.0700 | val_ROC-AUC=0.7071 | val_PR-AUC=0.0598 | valid_tags=186/188
baseline | Epoch 8/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 08 | time=99.5s | train_loss=0.0694 | val_ROC-AUC=0.7160 | val_PR-AUC=0.0635 | valid_tags=186/188
baseline | Epoch 9/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 09 | time=100.5s | train_loss=0.0689 | val_ROC-AUC=0.7215 | val_PR-AUC=0.0673 | valid_tags=186/188
baseline | Epoch 10/10:   0%|          | 0/250 [00:00<?, ?it/s]
baseline | epoch 10 | time=102.3s | train_loss=0.0685 | val_ROC-AUC=0.7433 | val_PR-AUC=0.0698 | valid_tags=186/188

BASELINE params
Trainable params: 87,550 / Total params: 87,550

BASELINE SUMMARY
TEST ROC-AUC (macro): 0.7263
TEST PR-AUC  (macro): 0.0708
TEST valid_tags: 188/188 (dropped=0)
Emissions (kg CO2eq): 0.0005499134254755918
Resources summary: {'wall_time_s': 1032.8490149974823, 'cpu_mean_pct': 29.770428015564203, 'cpu_max_pct': 93.1, 'ram_mean_mb': 420.8598005836576, 'ram_max_mb': 647.46875, 'n_samples': 1028, 'ac
cel_mem_mean_mb': 395.379299297407, 'accel_mem_max_mb': 683.9228515625}
```







FINAL TEST

ALL RESULTS + ADDITIONAL: CODECARBON PLUS TRACKED TRAINING + SUMMARY

In [55]:

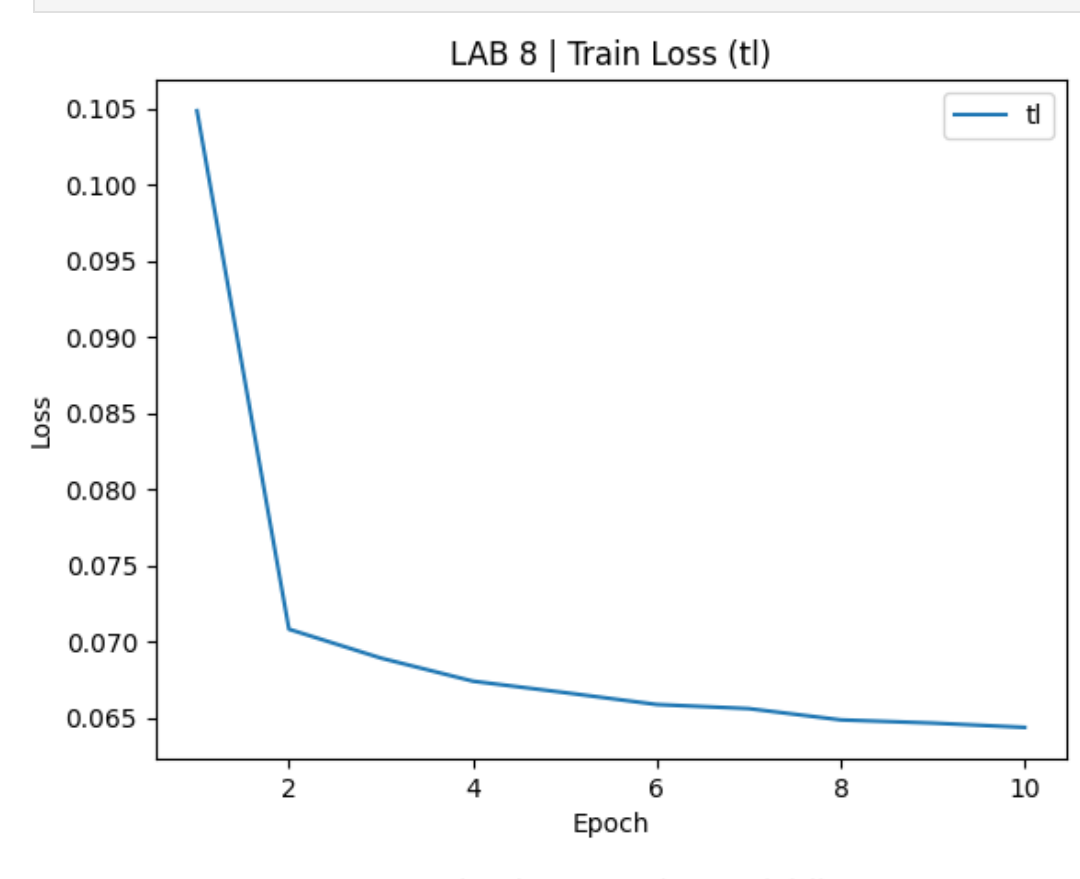
```
print(color.BOLD + "\nFINAL COMPARISON (TL vs BASELINE)" + color.END)
print(f"Device: {device}")
print(f"Train items: {len(train_df)} | Val items: {len(val_df)} | Test items: {len(test_df)}")
print(f"Tags/classes: {n_classes}")

for mode in ["tl", "baseline"]:
    res = results[mode]
    print(color.BOLD + f"\n{mode.upper()} FINAL" + color.END)
    print(f"TEST ROC-AUC (macro): {res['test_roc']:.4f}")
    print(f"TEST PR-AUC  (macro): {res['test_pr']:.4f}")
    print(f"TEST valid_tags: {res['test_diag']['valid_classes']}/{res['test_diag']['total_classes']} (dropped={res['test_diag']['dropped_classes']})")
    print("Emissions (kg CO2eq):", res["emissions_kg"])
    print("Resources summary:", res["resources"])
```

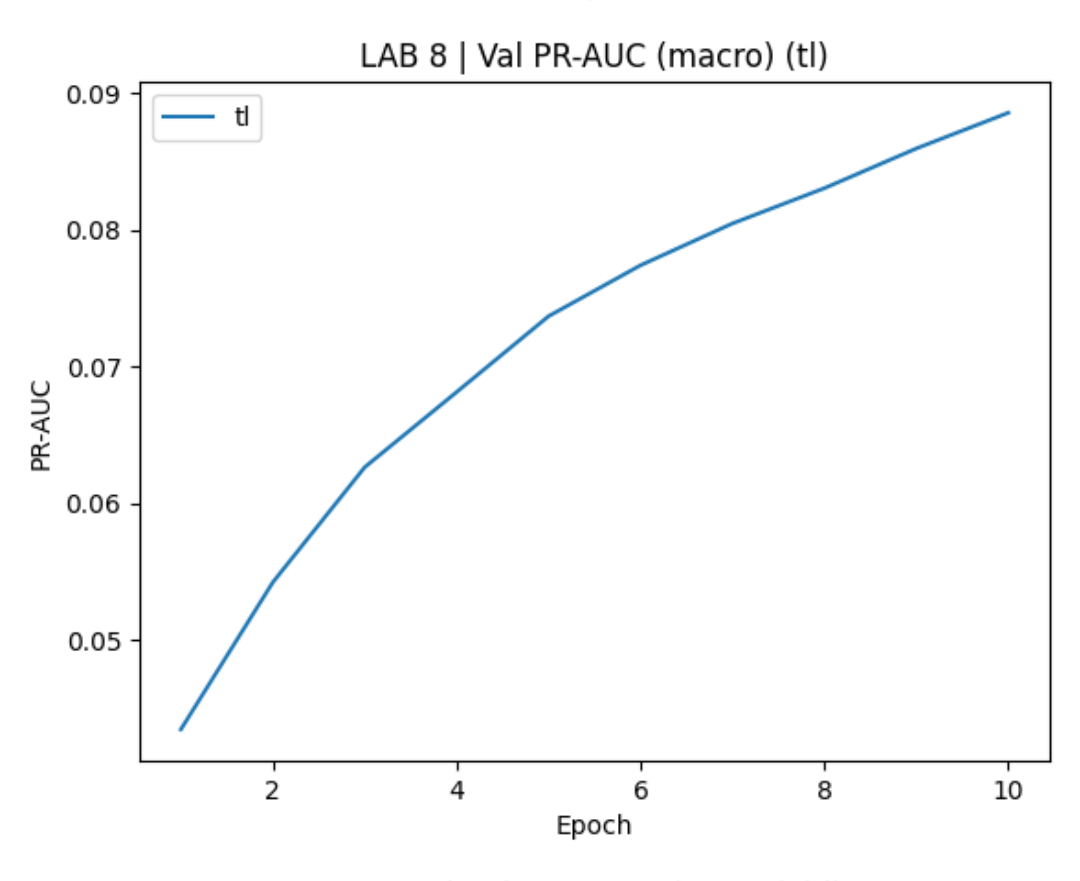
```
FINAL COMPARISON (TL vs BASELINE)
Device: mps
Train items: 8000 | Val items: 2000 | Test items: 2000
Tags/classes: 188

TL FINAL
TEST ROC-AUC (macro): 0.7621
TEST PR-AUC  (macro): 0.0868
TEST valid_tags: 188/188 (dropped=0)
Emissions (kg CO2eq): 0.001293008618316594
Resources summary: {'wall_time_s': 2430.143227815628, 'cpu_mean_pct': 23.911890999174236, 'cpu_max_pct': 90.1, 'ram_mean_mb': 815.8569170623451, 'ram_max_mb': 1159.28125, 'n_samples': 2422, 'ac
cel_mem_mean_mb': 406.30812724181976, 'accel_mem_max_mb': 785.61328125}

BASELINE FINAL
TEST ROC-AUC (macro): 0.7263
TEST PR-AUC  (macro): 0.0708
TEST valid_tags: 188/188 (dropped=0)
Emissions (kg CO2eq): 0.0005499134254755918
Resources summary: {'wall_time_s': 1032.8490149974823, 'cpu_mean_pct': 29.770428015564203, 'cpu_max_pct': 93.1, 'ram_mean_mb': 420.8598005836576, 'ram_max_mb': 647.46875, 'n_samples': 1028, 'ac
cel_mem_mean_mb': 395.379299297407, 'accel_mem_max_mb': 683.9228515625}
```

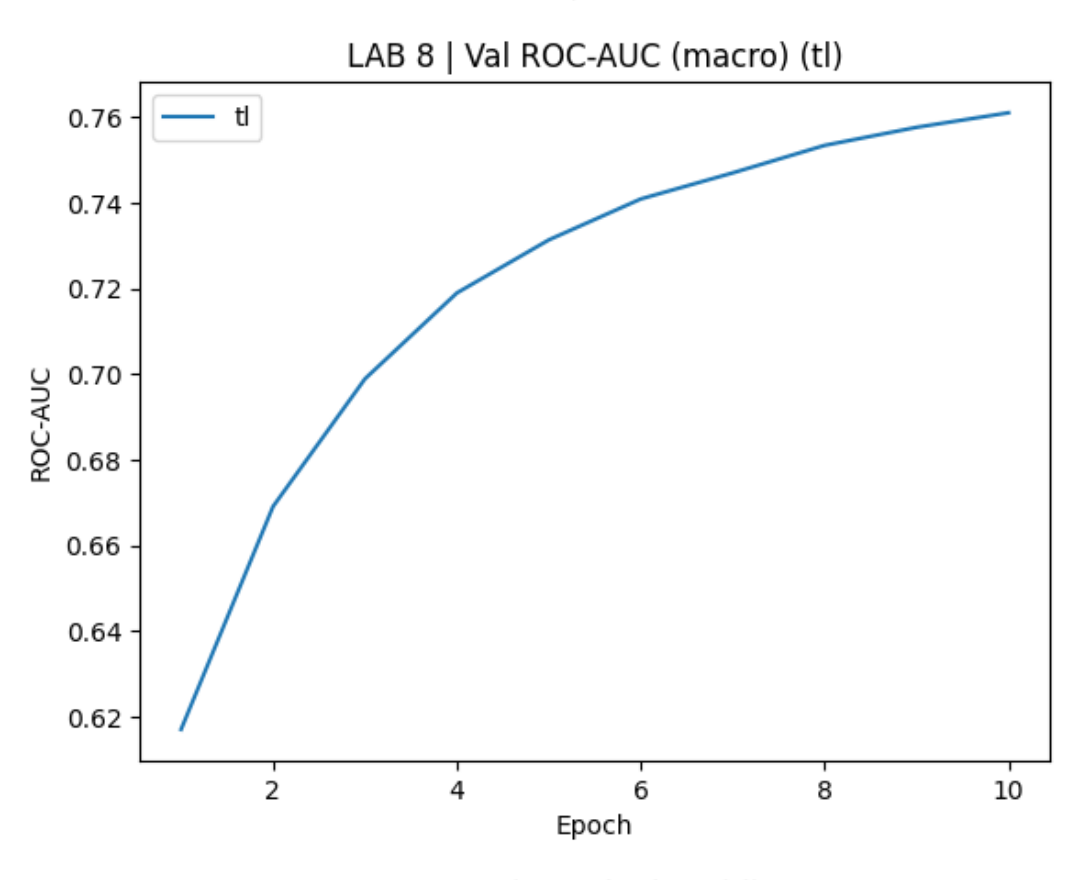
CHECKS

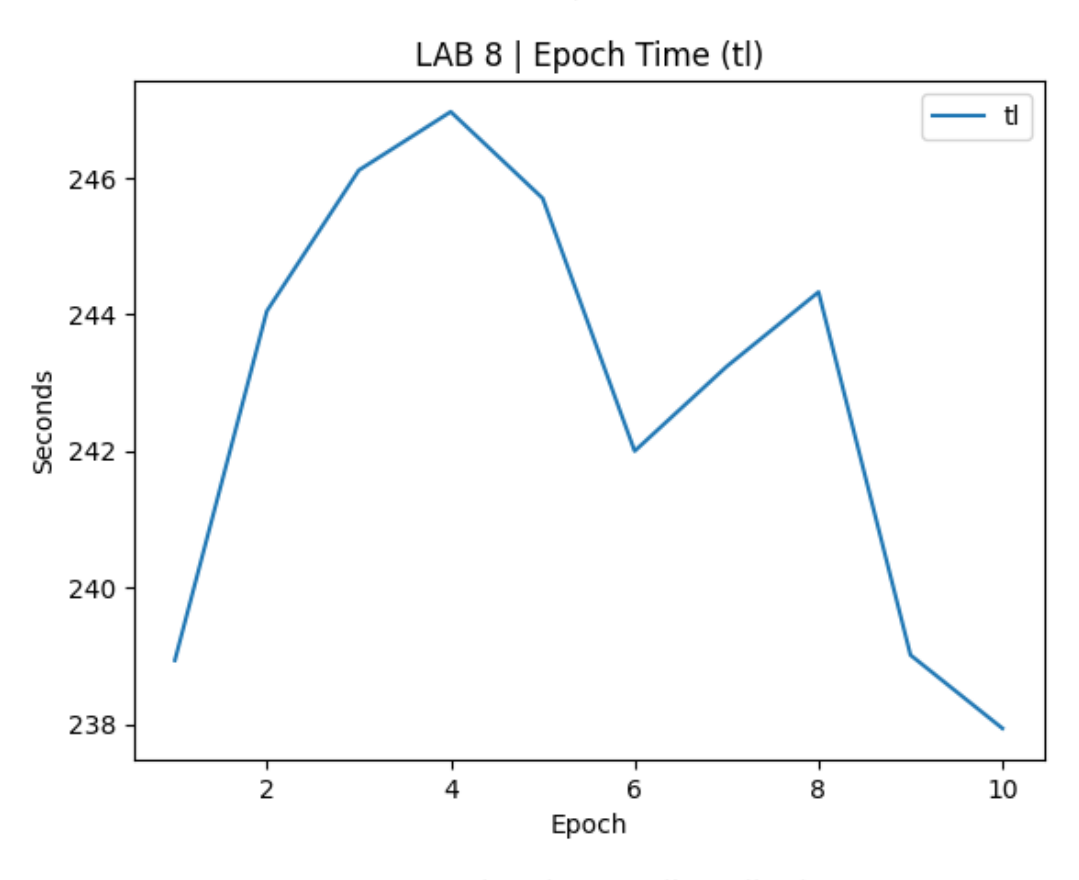
In [56]:

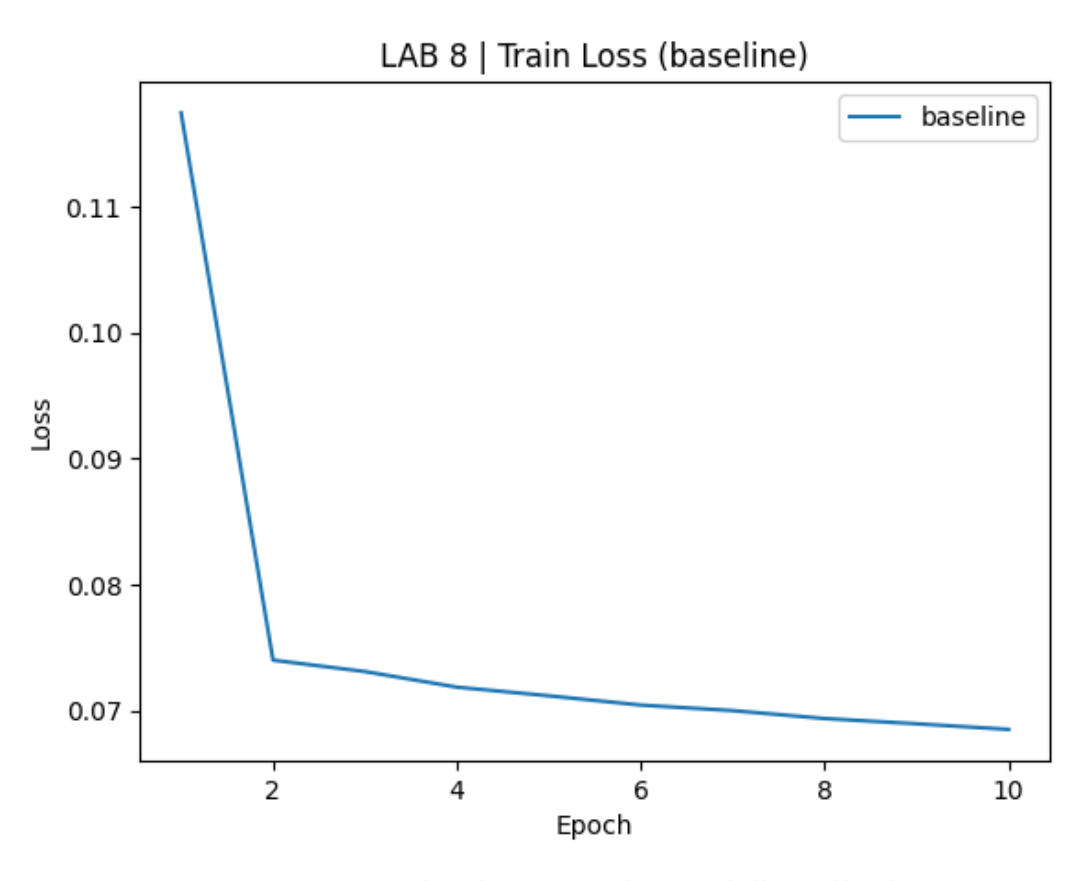
```
# SANITY CHECK 1: LEARNING CURVES (train loss / val PR-AUC / val ROC-AUC)
import pandas as pd
import matplotlib.pyplot as plt

def plot_history(results, title_prefix=""):
    for mode in ["tl", "baseline"]:
        if mode not in results:
            continue
        hist = results[mode]["history"]
        if not hist:
            print(f"No history for {mode}")
            continue

        df = pd.DataFrame(hist)

        plt.figure()
        plt.plot(df["epoch"], df["train_loss"], label=f"{mode}")
        plt.title(f"{title_prefix}Train Loss ({mode})")
        plt.xlabel("Epoch")
        plt.ylabel("Loss")
        plt.legend()
        plt.show()

        plt.figure()
        plt.plot(df["epoch"], df["val_pr"], label=f"{mode}")
        plt.title(f"{title_prefix}Val PR-AUC (macro) ({mode})")
        plt.xlabel("Epoch")
        plt.ylabel("PR-AUC")
        plt.legend()
        plt.show()

        plt.figure()
        plt.plot(df["epoch"], df["val_roc"], label=f"{mode}")
        plt.title(f"{title_prefix}Val ROC-AUC (macro) ({mode})")
        plt.xlabel("Epoch")
        plt.ylabel("ROC-AUC")
        plt.legend()
        plt.show()

        plt.figure()
        plt.plot(df["epoch"], df["epoch_time_s"], label=f"{mode}")
        plt.title(f"{title_prefix}Epoch Time ({mode})")
        plt.xlabel("Epoch")
        plt.ylabel("Seconds")
        plt.legend()
        plt.show()

plot_history(results, title_prefix="LAB 8 | ")
```

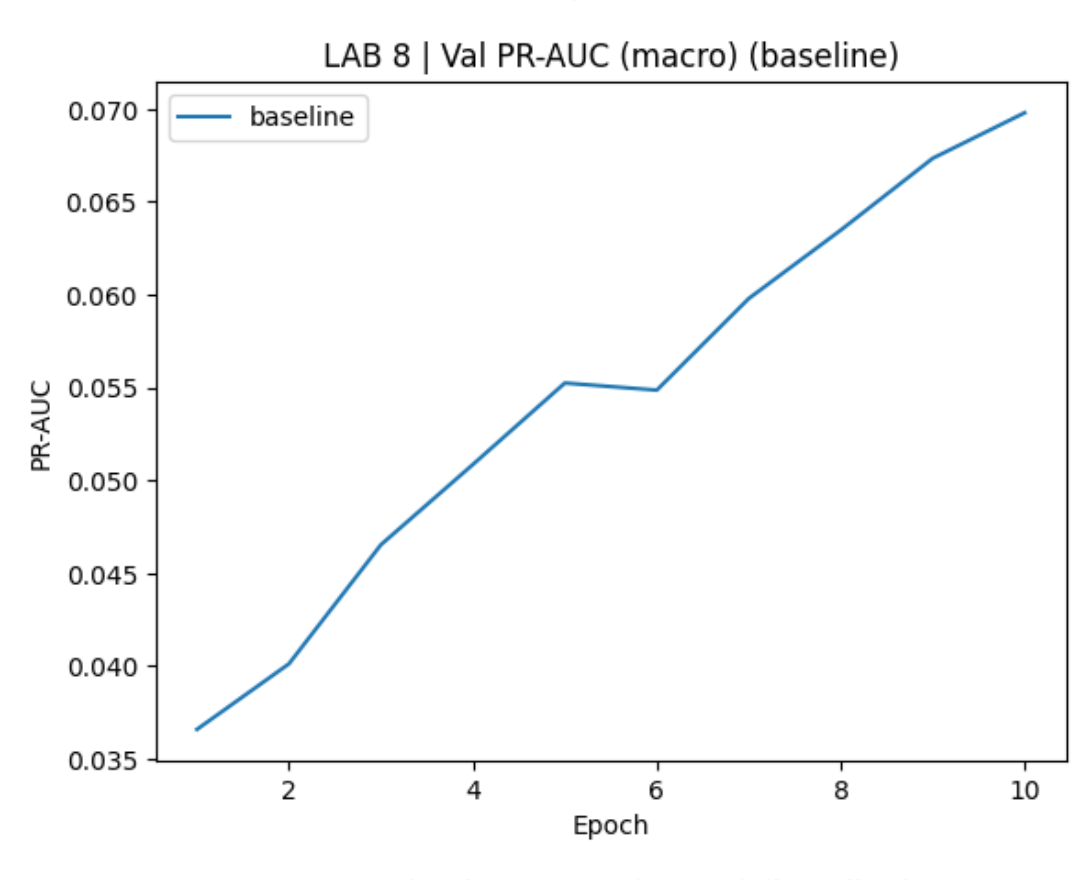


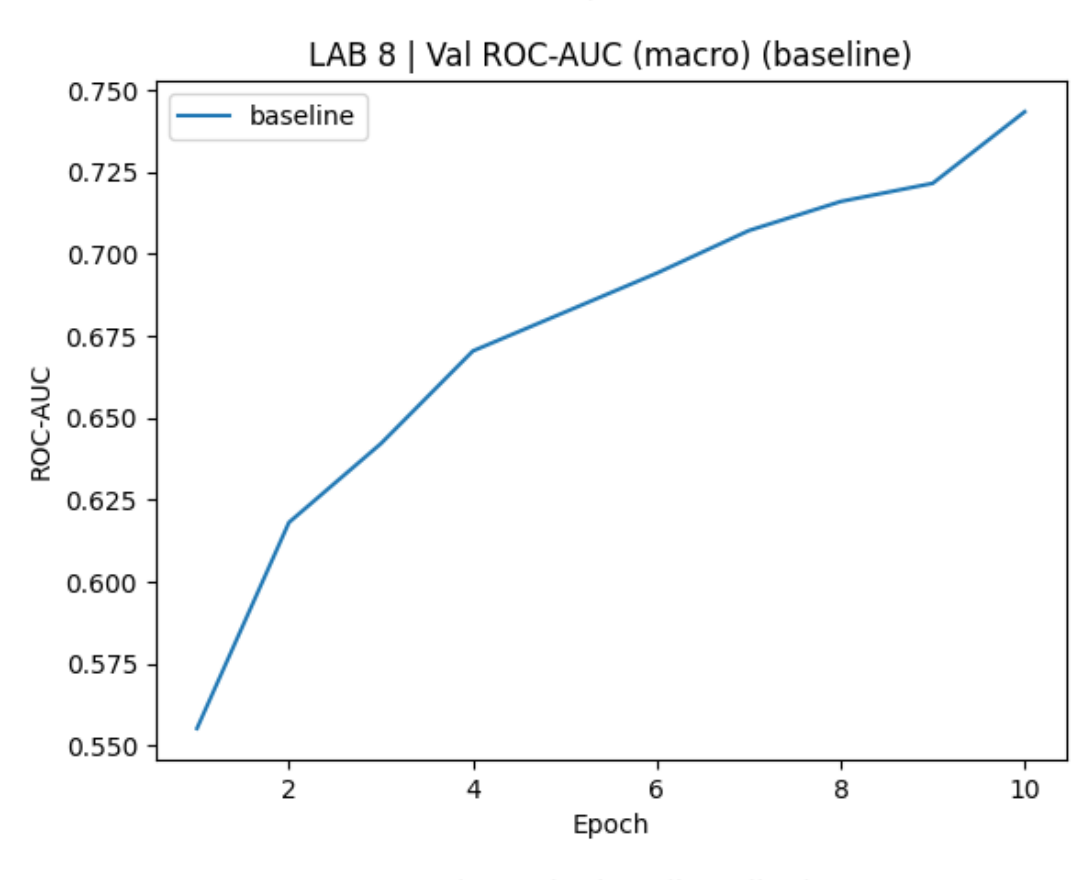


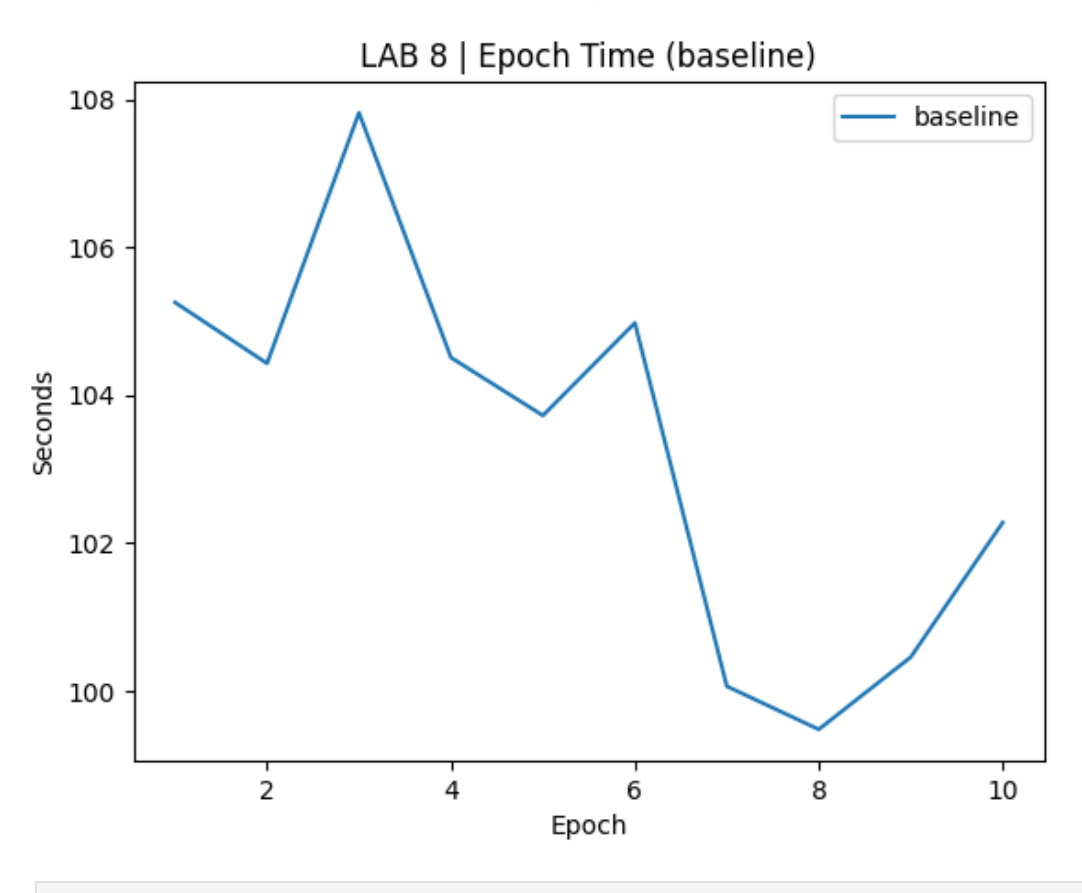












In [ ]:

```
# SANITY CHECK 2: INFERENCE (Top-K tags) + AUDIO + GROUND TRUTH (multi-label)
import numpy as np
import torch
import matplotlib.pyplot as plt

try:
    from IPython.display import Audio, display
except Exception:
    Audio = None
    display = None

def topk_tags(probs_1d, tag_cols, k=10):
    probs_1d = np.asarray(probs_1d).reshape(-1)
    idx = np.argsort(-probs_1d)[:k]
    return [(tag_cols[i], float(probs_1d[i])) for i in idx]

def infer_one(model, mode, x_1d_tensor):
    """
    x_1d_tensor: shape [T]
    returns probs: shape [n_classes]
    """
    model.eval()
    with torch.no_grad():
        x = x_1d_tensor.to(device)
        if x.dim() == 1:
            x = x.unsqueeze(0)

        if mode == "baseline":
            x_in = feature_extractor(x)
            # [B, n_mels, time]
            logits = model(x_in)
        else:
            logits = model(x)

        probs = torch.sigmoid(logits).squeeze(0).detach().cpu().numpy()
    return probs

def show_inference_on_sample(ds, idx, k=10):
    item = ds[int(idx)]
    x = item["audio"].detach().cpu()
    y = item["labels"].detach().cpu().numpy()

    # ground truth tags (where label==1)
    gt_idx = np.where(y > 0.5)[0].tolist()
    gt_tags = [TAG_COLS[i] for i in gt_idx]
    gt_tags = gt_tags[:30]
    # avoid huge prints

    print(color.BOLD + f"\nSample idx={idx} | split={getattr(ds, 'split', 'unknown')}" + color.END)
    print(f"Ground-truth tags (first 30): {gt_tags}")

    # audio
    if Audio is not None and display is not None:
        display(Audio(x.numpy(), rate=CFG["sample_rate"]))

    # predictions
    for mode in ["tl", "baseline"]:
        probs = infer_one(trained_models[mode], mode, x)
        tk = topk_tags(probs, TAG_COLS, k=k)
        print(color.BOLD + f"\n{mode.upper()} Top-{k} predicted tags:" + color.END)
        for t, p in tk:
            print(f"  {t:20s}  p={p:.3f}")

# run on a few random test samples
rng = np.random.RandomState(SEED)
for _ in range(3):
    ridx = int(rng.randint(0, len(test_dataset)))
    show_inference_on_sample(test_dataset, ridx, k=10)
```

```
Sample idx=1126 | split=test
Ground-truth tags (first 30): ['rock']
▶ 0:00 0:02

TL Top-10 predicted tags:
  rock                  p=0.836
  metal                 p=0.488
  loud                  p=0.432
  hard rock             p=0.348
  guitar                p=0.290
  fast                  p=0.154
  heavy                 p=0.139
  drums                 p=0.136
  male                  p=0.116
  male vocal            p=0.105

BASELINE Top-10 predicted tags:
  guitar                p=0.442
  rock                  p=0.285
  slow                  p=0.182
  vocal                 p=0.176
  male                  p=0.173
  ambient               p=0.157
  loud                  p=0.143
  piano                 p=0.142
  singing               p=0.142
  male vocal            p=0.139

Sample idx=1459 | split=test
Ground-truth tags (first 30): []
▶ 0:00 0:02

TL Top-10 predicted tags:
  guitar                p=0.184
  drums                 p=0.129
  beat                  p=0.120
  slow                  p=0.089
  techno                p=0.078
  electronic            p=0.068
  fast                  p=0.053
  synth                 p=0.051
  vocal                 p=0.046
  female                p=0.040
```

```
BASELINE Top-10 predicted tags:
  techno                p=0.488
  electronic            p=0.413
  beat                  p=0.329
  drums                 p=0.294
  beats                 p=0.227
  synth                 p=0.168
  fast                  p=0.160
  guitar                p=0.137
  drum                  p=0.119
  dance                 p=0.090

Sample idx=860 | split=test
Ground-truth tags (first 30): ['no vocal', 'piano']
```

▶ 0:00 0:02

```
TL Top-10 predicted tags:
  slow                  p=0.412
  quiet                 p=0.332
  piano                 p=0.292
  guitar                p=0.181
  soft                  p=0.135
  ambient               p=0.091
  silence               p=0.060
  harp                  p=0.050
  classical             p=0.048
  low                   p=0.032

BASELINE Top-10 predicted tags:
  techno                p=0.336
  electronic            p=0.305
  beat                  p=0.250
  guitar                p=0.195
  drums                 p=0.169
  slow                  p=0.163
  synth                 p=0.147
  bass                  p=0.135
  beats                 p=0.127
  quiet                 p=0.095
```

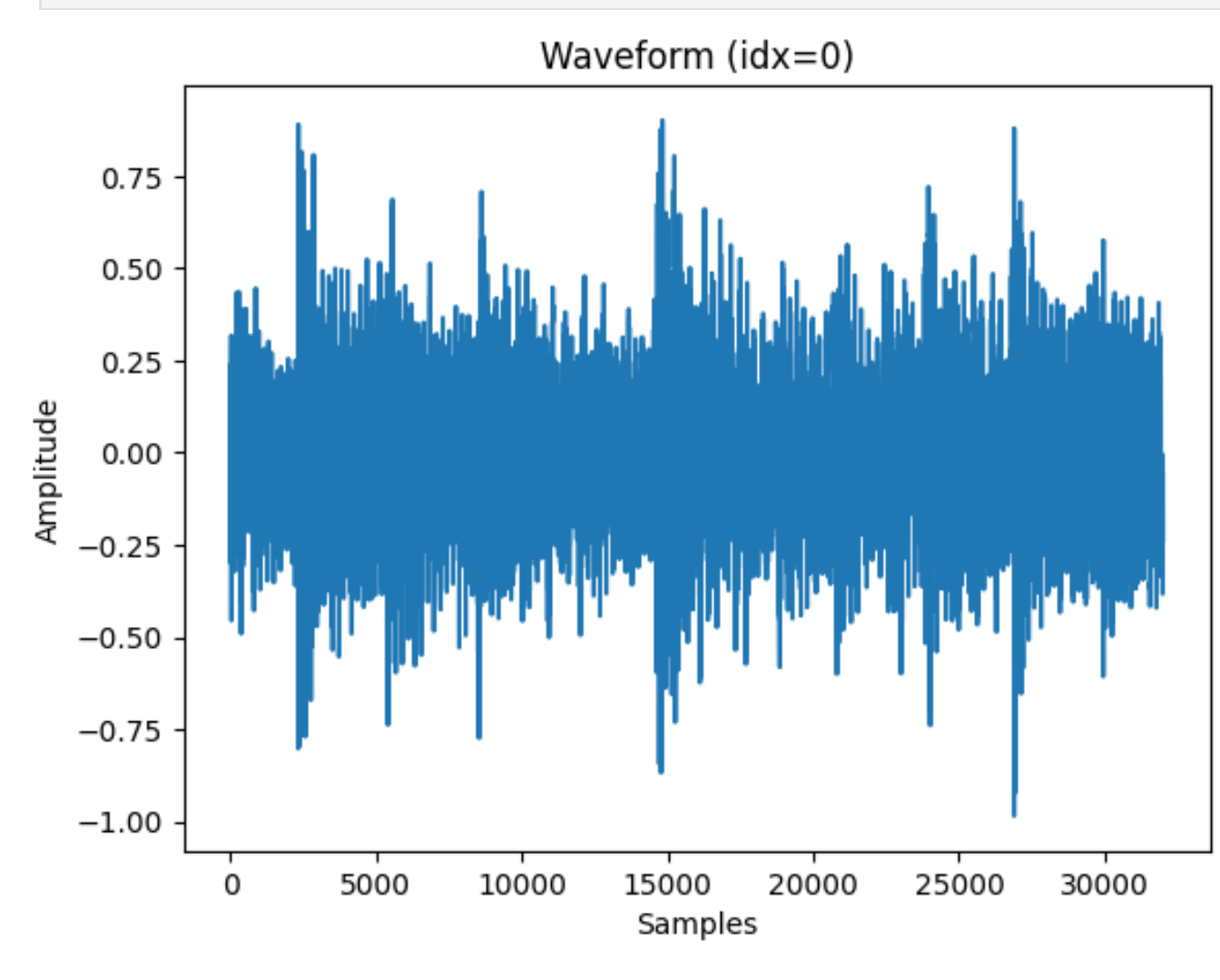
In [58]:

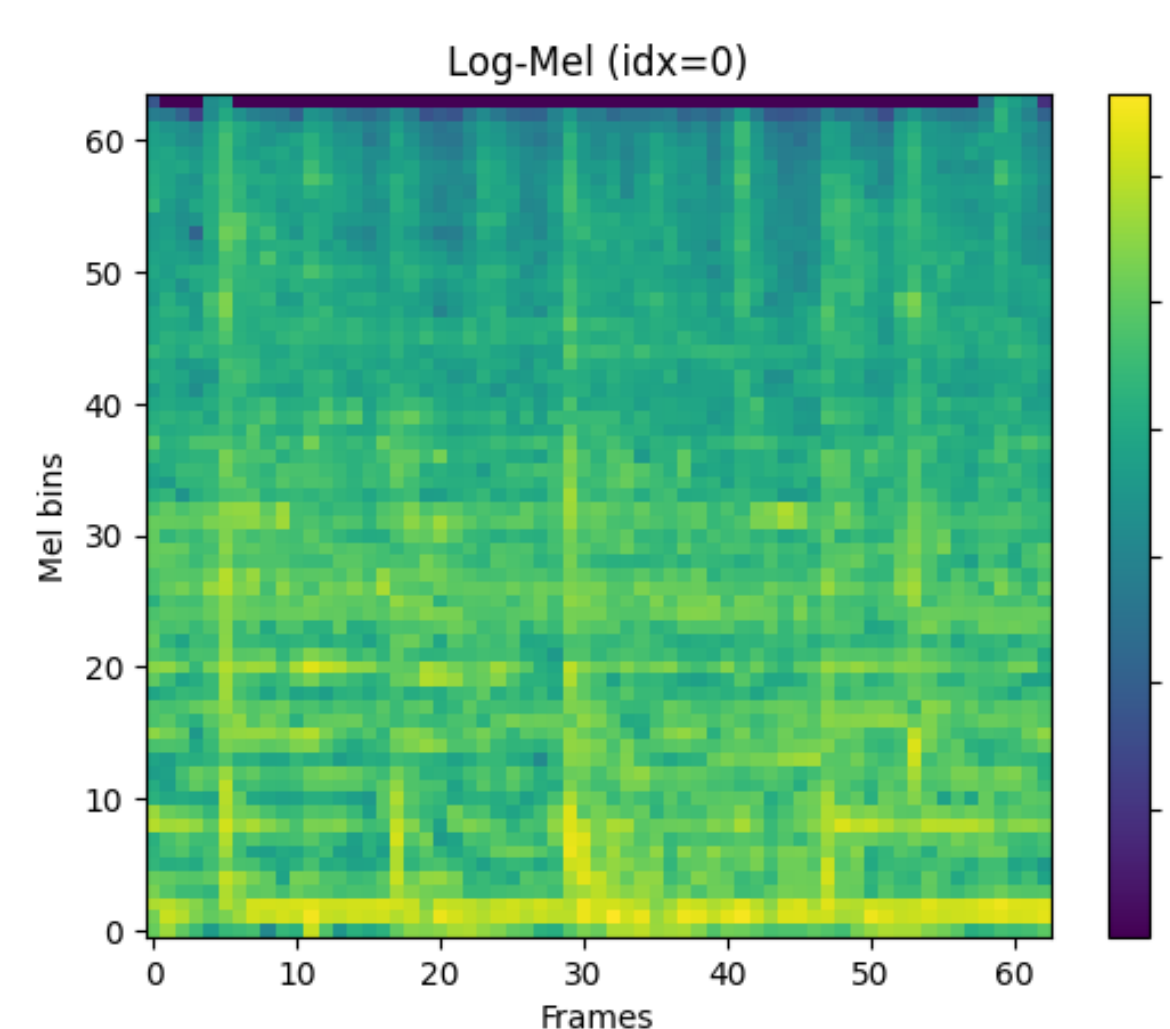
```python
# SANITY CHECK 3: TL vs BASELINE AGREEMENT (Top-K overlap / Jaccard)
import numpy as np

def jaccard(a, b):
    a, b = set(a), set(b)
    return len(a & b) / max(1, len(a | b))

def compare_models_on_sample(ds, idx, k=10):
    item = ds[int(idx)]
    x = item["audio"].detach().cpu()

    probs_tl = infer_one(trained_models["tl"], "tl", x)
    probs_bl = infer_one(trained_models["baseline"], "baseline", x)

    top_tl = [t for t, _ in topk_tags(probs_tl, TAG_COLS, k=k)]
    top_bl = [t for t, _ in topk_tags(probs_bl, TAG_COLS, k=k)]

    jac = jaccard(top_tl, top_bl)

    print(color.BOLD + f"\nCOMPARE idx={idx} | Top-{k} Jaccard={jac:.3f}" + color.END)
    print("TL top-k:", top_tl)
    print("BL top-k:", top_bl)

# compare on a few samples
for idx in [0, len(test_dataset)//3, (2*len(test_dataset))//3]:
    compare_models_on_sample(test_dataset, idx, k=10)
```

```
COMPARE idx=0 | Top-10 Jaccard=0.111
TL top-k: ['loud', 'techno', 'rock', 'synth', 'ambient', 'electronic', 'fast', 'metal', 'dance', 'drums']
BL top-k: ['guitar', 'ambient', 'slow', 'rock', 'vocal', 'piano', 'vocals', 'male', 'singing', 'male vocal']

COMPARE idx=666 | Top-10 Jaccard=0.250
TL top-k: ['guitar', 'rock', 'fast', 'drums', 'strings', 'classical', 'violin', 'country', 'male vocal', 'slow']
BL top-k: ['techno', 'electronic', 'drums', 'guitar', 'beat', 'synth', 'fast', 'beats', 'slow', 'drum']

COMPARE idx=1333 | Top-10 Jaccard=0.538
TL top-k: ['ambient', 'drums', 'synth', 'guitar', 'electronic', 'slow', 'fast', 'no vocals', 'flute', 'techno']
BL top-k: ['electronic', 'techno', 'drums', 'guitar', 'beat', 'synth', 'beats', 'fast', 'slow', 'drum']
```

In [59]:

```python
# SANITY CHECK 4: FEATURE VISUALIZATION (Waveform + Log-Mel) on a test sample
import matplotlib.pyplot as plt
import torch

def show_features(ds, idx):
    item = ds[int(idx)]
    x = item["audio"].detach().cpu()  # [T]

    plt.figure()
    plt.plot(x.numpy())
    plt.title(f"Waveform (idx={idx})")
    plt.xlabel("Samples")
    plt.ylabel("Amplitude")
    plt.show()

    with torch.no_grad():
        mel = feature_extractor(x.unsqueeze(0).to(device)).squeeze(0).detach().cpu().numpy()

    plt.figure()
    plt.imshow(mel, aspect="auto", origin="lower")
    plt.title(f"Log-Mel (idx={idx})")
    plt.xlabel("Frames")
    plt.ylabel("Mel bins")
    plt.colorbar()
    plt.show()

show_features(test_dataset, idx=0)
```

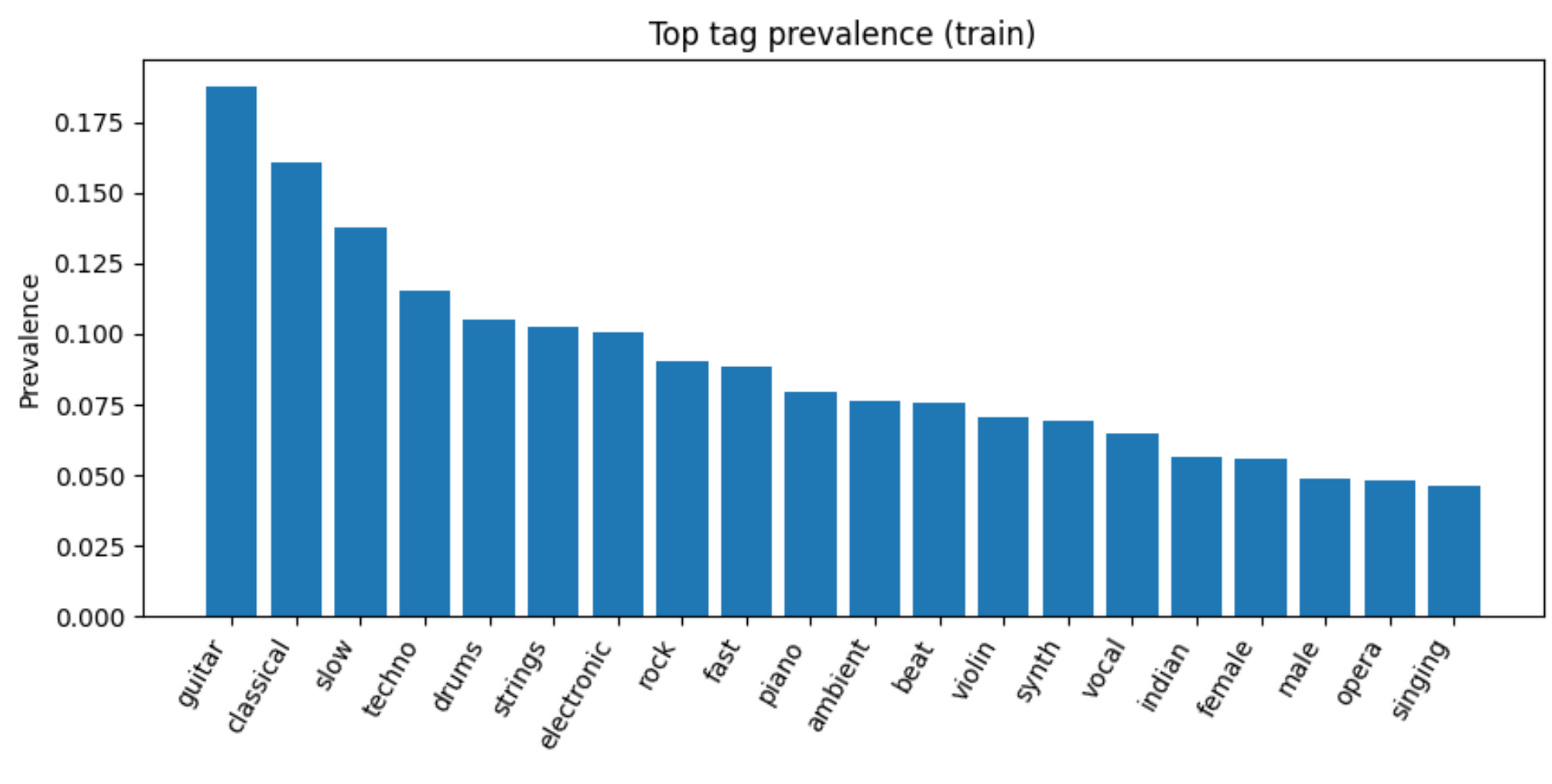




In [ ]:

```python
# SANITY CHECK 5: LABEL DISTRIBUTION (prevalence of tags) in train/val/test
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

def tag_prevalence(df_split, name="split"):
    # df_split has clip_id, mp3_path
    clip_ids = df_split["clip_id"].values
    Y = ann_idx.loc[clip_ids, TAG_COLS].values.astype(np.float32)
    # [N, C]
    prev = Y.mean(axis=0)
    # prevalence per tag

    out = pd.DataFrame({"tag": TAG_COLS, "prevalence": prev}).sort_values("prevalence", ascending=False).reset_index(drop=True)
    print(color.BOLD + f"\n{name} | N={len(df_split)} | mean positive tags/sample={Y.sum(axis=1).mean():.2f}" + color.END)
    return out

prev_train = tag_prevalence(train_df, "train")
prev_val   = tag_prevalence(val_df, "val")
prev_test  = tag_prevalence(test_df, "test")

# plot top 20 tags of train
topN = 20
plt.figure(figsize=(10,4))
plt.bar(prev_train["tag"].head(topN), prev_train["prevalence"].head(topN))
plt.title("Top tag prevalence (train)")
plt.xticks(rotation=60, ha="right")
plt.ylabel("Prevalence")
plt.show()
```

```
train | N=8000 | mean positive tags/sample=3.45

val | N=2000 | mean positive tags/sample=3.38

test | N=2000 | mean positive tags/sample=3.41
```



In [61]:

```python
# SANITY CHECK 6: PREDICTION CONFIDENCE DIAGNOSTIC (are outputs non-trivial?)
import numpy as np

def confidence_stats(model, mode, ds, n=50):
    rng = np.random.RandomState(SEED)
    max_ps, mean_ps = [], []
    for _ in range(n):
        idx = int(rng.randint(0, len(ds)))
        x = ds[idx]["audio"].detach().cpu()
        probs = infer_one(model, mode, x)
        max_ps.append(float(np.max(probs)))
        mean_ps.append(float(np.mean(probs)))
    return {
        "max_p_mean": float(np.mean(max_ps)),
        "max_p_std": float(np.std(max_ps)),
        "mean_p_mean": float(np.mean(mean_ps)),
        "mean_p_std": float(np.std(mean_ps)),
    }

for mode in ["tl", "baseline"]:
    stats = confidence_stats(trained_models[mode], mode, test_dataset, n=80)
    print(color.BOLD + f"\n{mode.upper()} confidence stats on test:" + color.END)
    for k, v in stats.items():
        print(f"  {k}: {v:.4f}")
```

```
TL confidence stats on test:
  max_p_mean: 0.4318
  max_p_std: 0.1830
  mean_p_mean: 0.0188
  mean_p_std: 0.0052

BASELINE confidence stats on test:
  max_p_mean: 0.3822
  max_p_std: 0.0953
  mean_p_mean: 0.0235
  mean_p_std: 0.0039
```

In [ ]:

```python
import pandas as pd

def make_summary_table(results, device, train_df, val_df, test_df, n_classes):
    rows = []
    for mode in ["tl", "baseline"]:
        r = results[mode]
        # última fila de history (o mejor_epoch if saved)
        last = r["history"][-1] if r["history"] else {}
        rows.append({
            "mode": mode,
            "device": device,
            "train_items": len(train_df),
            "val_items": len(val_df),
            "test_items": len(test_df),
            "tags": n_classes,
            "test_roc_macro": r["test_roc"],
            "test_pr_macro": r["test_pr"],
            "emissions_kg": r["emissions_kg"],
            "wall_time_s": r["resources"].get("wall_time_s", None),
            "cpu_mean_pct": r["resources"].get("cpu_mean_pct", None),
            "ram_max_mb": r["resources"].get("ram_max_mb", None),
            "accel_mem_max_mb": r["resources"].get("accel_mem_max_mb", None),
            "last_train_loss": last.get("train_loss", None),
            "last_val_roc": last.get("val_roc", None),
            "last_val_pr": last.get("val_pr", None),
        })
    return pd.DataFrame(rows)

summary_df = make_summary_table(results, device, train_df, val_df, test_df, n_classes)
summary_df
```

Out[ ]:

	mode	device	train_items	val_items	test_items	tags	test_roc_macro	test_pr_macro	emissions_kg	wall_time_s	cpu_mean_pct	ram_max_mb	accel_mem_max_mb	last_train_loss	last_val_roc	last_val_pr
0	tl	mps	8000	2000	2000	188	0.762073	0.08678	0.001293	2430.143228	23.911891	1159.28125	785.613281	0.064370	0.761027	0.088554
1	baseline	mps	8000	2000	2000	188	0.726340	0.07084	0.000550	1032.849015	29.770428	647.46875	683.922852	0.068493	0.743335	0.069793

In [63]:

```python
history_df = pd.DataFrame(results["baseline"]["history"])   # o "tl"
history_df.tail(10)
```

Out[63]:

	epoch	train_loss	val_roc	val_pr	epoch_time_s	val_valid_tags	val_total_tags	val_dropped_tags
0	1	0.117454	0.555201	0.036599	105.251535	186	188	2
1	2	0.073993	0.618042	0.040129	104.427825	186	188	2
2	3	0.073084	0.642090	0.046538	107.816116	186	188	2
3	4	0.071847	0.670350	0.050858	104.506716	186	188	2
4	5	0.071158	0.682197	0.055245	103.723573	186	188	2
5	6	0.070425	0.694096	0.054862	104.975005	186	188	2
6	7	0.069984	0.707079	0.059793	100.062536	186	188	2
7	8	0.069356	0.715966	0.063475	99.479908	186	188	2
8	9	0.068939	0.721506	0.067341	100.456729	186	188	2
9	10	0.068493	0.743335	0.069793	102.275962	186	188	2

In [64]:

```python
summary_df.style.format({
    "test_roc_macro": "{:.4f}",
    "test_pr_macro": "{:.4f}",
    "emissions_kg": "{:.6f}",
    "wall_time_s": "{:.1f}",
    "cpu_mean_pct": "{:.1f}",
    "ram_max_mb": "{:.1f}",
    "accel_mem_max_mb": "{:.1f}",
    "last_train_loss": "{:.4f}",
    "last_val_roc": "{:.4f}",
    "last_val_pr": "{:.4f}",
})
```

Out[64]:

	mode	device	train_items	val_items	test_items	tags	test_roc_macro	test_pr_macro	emissions_kg	wall_time_s	cpu_mean_pct	ram_max_mb	accel_mem_max_mb	last_train_loss	last_val_roc	last_val_pr
0	tl	mps	8000	2000	2000	188	0.7621	0.0868	0.001293	2430.1	23.9	1159.3	785.6	0.0644	0.7610	0.0886
1	baseline	mps	8000	2000	2000	188	0.7263	0.0708	0.000550	1032.8	29.8	647.5	683.9	0.0685	0.7433	0.0698

REPORT

INTRODUCTION

The goal of the laboratory is to study the use of architectures based on transformers for multilabel music tagging, in this case with MagnaTagATune dataset.

In this notebook there are two approaches taken:

1. A baseline transformer model trained from scratch on log mel specgtrograms
2. A transfer learning TL approach using pretrained wav2vec2 backbone, frozen and adapted with a lightweight classification head. This in order to have comparison points of the models.

The evalutation's focuse is not only on predictive performance but also on computational cost, resource usage and environmental impact. This is in order to align with the best practices in sustainable machine learning.

SETUP & ENVIRONMENT

The design of the notebook is to be self contained and portable. It can run locally and on Google Colab. In MacOS it profits on MPS.

The setup section:

- Installs dependencies automatically when in Colab.
- Detects the available compute backend (CUDA, MPS or CPU). Also based on previously made tests and labs.
- Fixes random seeds to ensure reproducibility.
- Adapts DataLoader params deending on the execution environment to avoid deadlocks and memory issues.

All the experiments reported below were run in a fixed evironement with a fixed random see and identical data splits to ensure a fair comparison between models.

DATASET: MAGNATAGATUNE

MTAT dataset is used. This works as a standard benchmark for the music taggin exercise containing shord audio clips annotated with multiple semantic tags.

Key characteristics:

- Audio clips are resampled to 16khz and truncated or paddded to 2seconds.
- each clip can have multiple active tags (multilabel)
- A total of 188 tags are considered.

The dataset is split into train, validation and tests with a 80-10-10 ratio.

!!To reduce computational cost, subsets of the orginal splits are optionally used while aiming to preserve label diversity.

DATA LOADING AND ROBUSTNESS

A custom dataset class is implemented to load the MTAT audio files robustly.

Design choices:

- Automatic discovery of the dataset root. This allows flexible folder layouts.
- Support for both torchaudio and ffmpeg decoding, ensuring compatibility with problematic MP3 files as MTAT has a lot of those.
- On the fly padding or random cropping during training. This is done to enforce fixed lengght inputs.
- Retry of the logic to prevent training crashes due to corrupted files.

These measures are implemented to prevend training crashes due to corrupted files.

FEATURE REPRESENTATION

For the baseline model audio signals are converted into log mel spectrograms. These are widely used in MIR tasks.

The feature extractore:

- Computes mel specrtograms using fixed STFT parameters
- Applies logarithmic compressin to better match "human perception"
- Is implemented as a PyTorch module for seamless integration into the training pipeline.

For the transfer learning approach, rw waveforms are directly fed into the pretrained wav2vec2 model. This leverages learned representations from large scale audio pretraining.

MODELS

Baseline transformer

The baseline model is a transformer encoder that operates on log mel spectrkgram "patches". It includes:

- Convolutional patchin layer
- Sinusoidal positional encodings along time and frequency
- Multiple transformer encoder blocks
- A hybrid pooling strategy that combines a CLS token and mean pooling.

This model was trained end to end on MTAT without external pretraining.

Transfer learning

The transfer learning model uses a pre trained wav2vec2 backbone. This is frozen during trianing. Only a lightweight linear classification head is trained on top of the extracted representations.

This approach tests whether representations learned from a large scale audio pretraining transfer effectively to music tagging with limited task specific training.

TRANING PROCEDURE

Both models were trained using:

- Binary cross-entropy loss with logits suitable for multilabel classsification.
- The Adam optimizer with a fixed learning rate.

During training, the following is logged:

- Training loss per epoch
- Validation ROC-AUC and PR-AUC
- Epoch duration and system resource usage

The best model checkpoint (based on the validation PR-AUC) is restored before the final evaluation.

EVALUATION METRICS

An important point is performance, so it is evaluated using:

- ROC-AUC (macro): this measures ranking quality across all the available tags.
- PR-AUC (micro): which is more informative under class imablance. This imbalance is sever in mtat.

Tags that are constant, either active or inactive, within a split are automatically excluded from metric computation to avoid invalid/non reliable scores.

RESOURCE MONITORING AND ENVIRONMENTAL IMPACT

In addition to performance, the following is tracked:

- CPU utilization and peak usage
- RAM and accelerator (GPU/MPS) memory consumption
- Total wall clock training time
- Estimated CO2 emissions using codecarbon.

This allows a direct comparison between models in terms of efficiency and sustainability. This highlights as well the trade offs between transfer learning and training from scratch.

RESULTS AND COMPARISON

The transfer learning model outperforms the baseline transformer in a consisten way, along all the runs and tests that were made for this lab. In both ROC-AUC and PR-AUC this is observable.

While the TL approach requires significantly more parameters, the number of trainable parameters is way smaller due to freezing backbone. This leads to an improved generalization at the cost of increased computation and energy consumption.

The baseline model is faster and more efficient in terms of energy but achieves a lower tagging performance. At the end, the decision of which model is better will be based on cost-time-scope.

SANITY CHECKS AND QUALITATIVE ANALYSIS

In order to verify that the models learn meaningful representations several sanity checks were performed:

1. That learning curves show stable convergence.
2. A qualitative inspection of the TOP K predicted tags for test samples.
3. Audio playback alongside ground truth and predicted tags.
4. Comparison between baseline and TL predictions using a Jaccard similarity.
5. Visualizations of the waveforms and log mel spectrograms.
6. An analysis of prediction confidence distributions.

These checks confirm that the models are stable, do not collapse to trivial solutions and important: they produce semantically plausible predictions.

DISCUSSION

The results demonstrate the effectiveness of transfer learning for music tagging, even when the pre trained model could have not been trained on music data but other kind of data.

The TL model achieves a high performance with fewer trainable parameters. This illustrates the benefit of reusable audio representations. However, important to note: this comes at the cost of increased computational and environmental impact.

These findings highlight an important trade off between accuracy and efficiency. I believe this must be considered when deploying large scale audio models as they can be a constraint in commercial environments or even a stopper in terms or geopolitical context.

CONCLUSION

The laboratory explored models of transformers for multilabel music tagging and compared training from scratch with a transfer learning approach.

The experiments done show:

- Transfer learning with wav2vec2 significantly improves tagging performance.
- Baseline transformers remain competitive, specially when computational resources and constraints are critical or important.
- Resource and emissions tracking provide a valuable insight beyond the accuracy metrics and give us an insight on how in large scale these models could be profitable, manageable, be ethic or easily implemented in different geopolitical contexts.

Overall, the notebook demonstrates a complete, reproducible and scalable workflow for modern audio ML experimentation.